Developing acoustic localization for wildlife in environmental impact assessment

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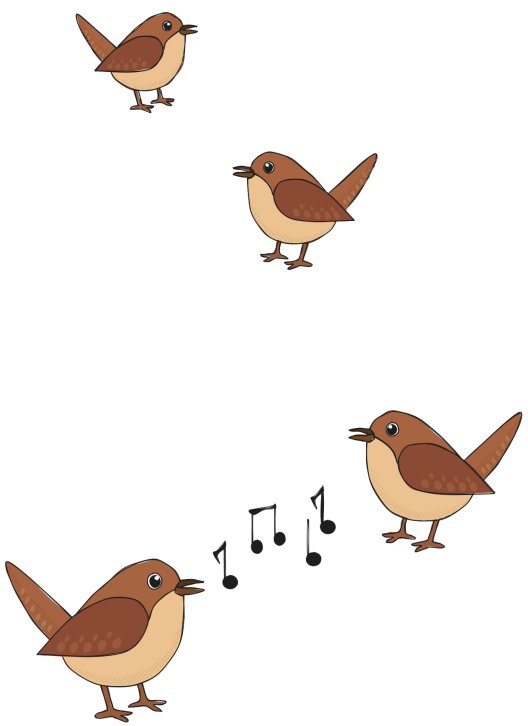
Assessing impacts

Challenges

Challenges



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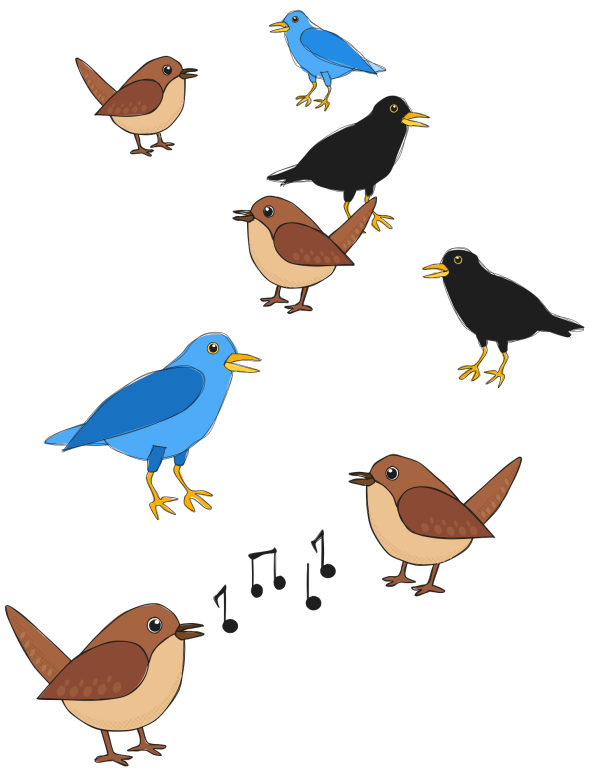

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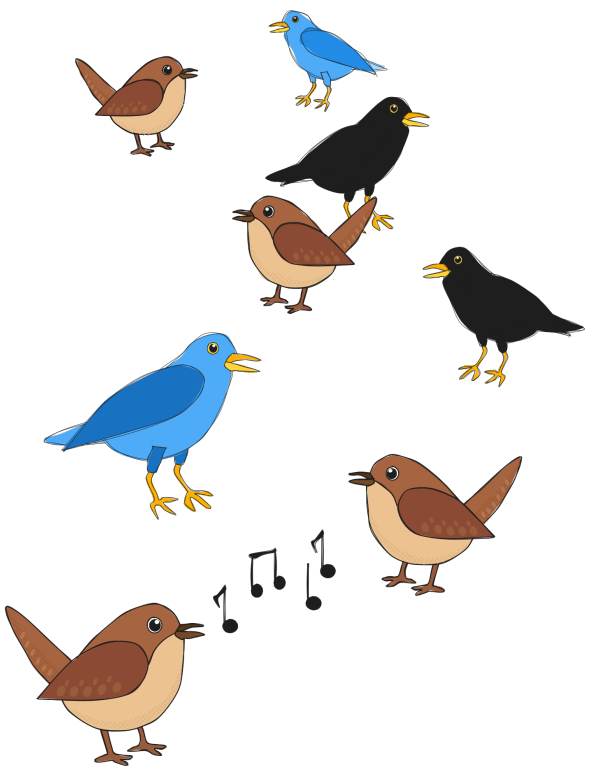

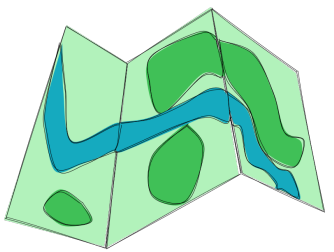

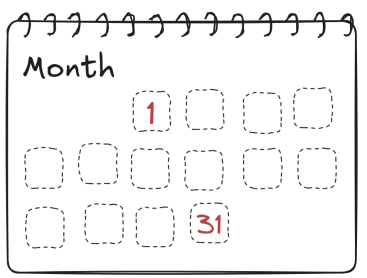

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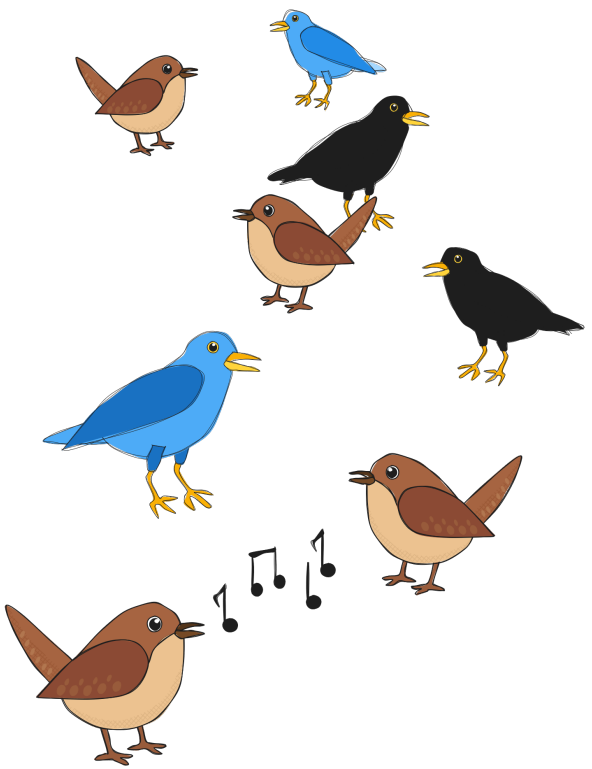

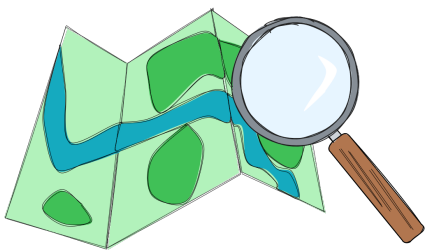

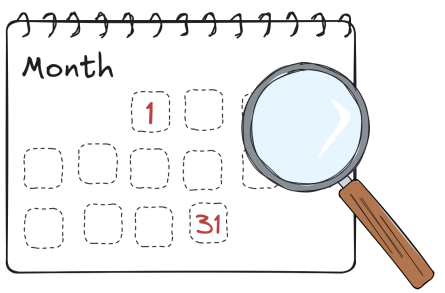

Behavior

Mennill et al. (2006), Dutilleux, Sandercock, and Kålås (2023), Teixeira, Maron, and Rensburg (2019)

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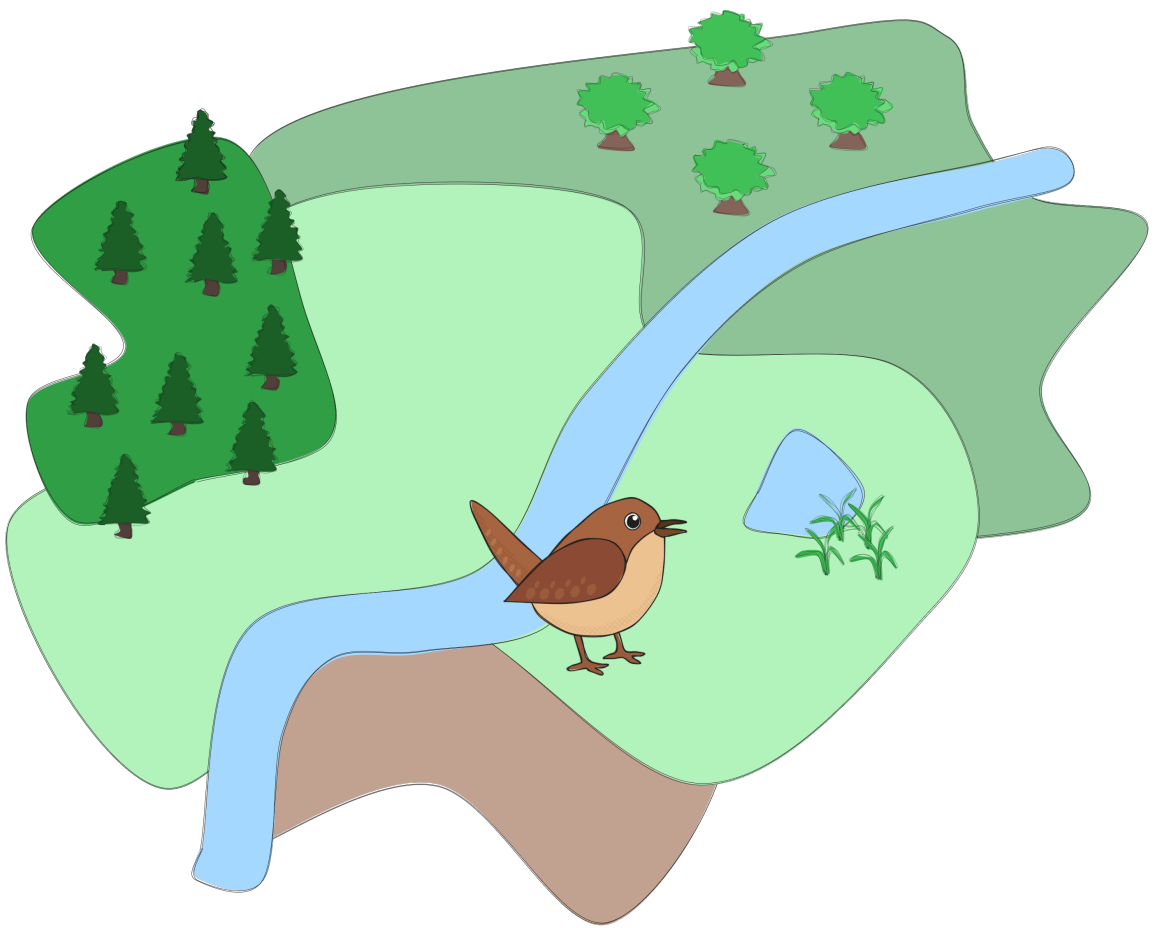

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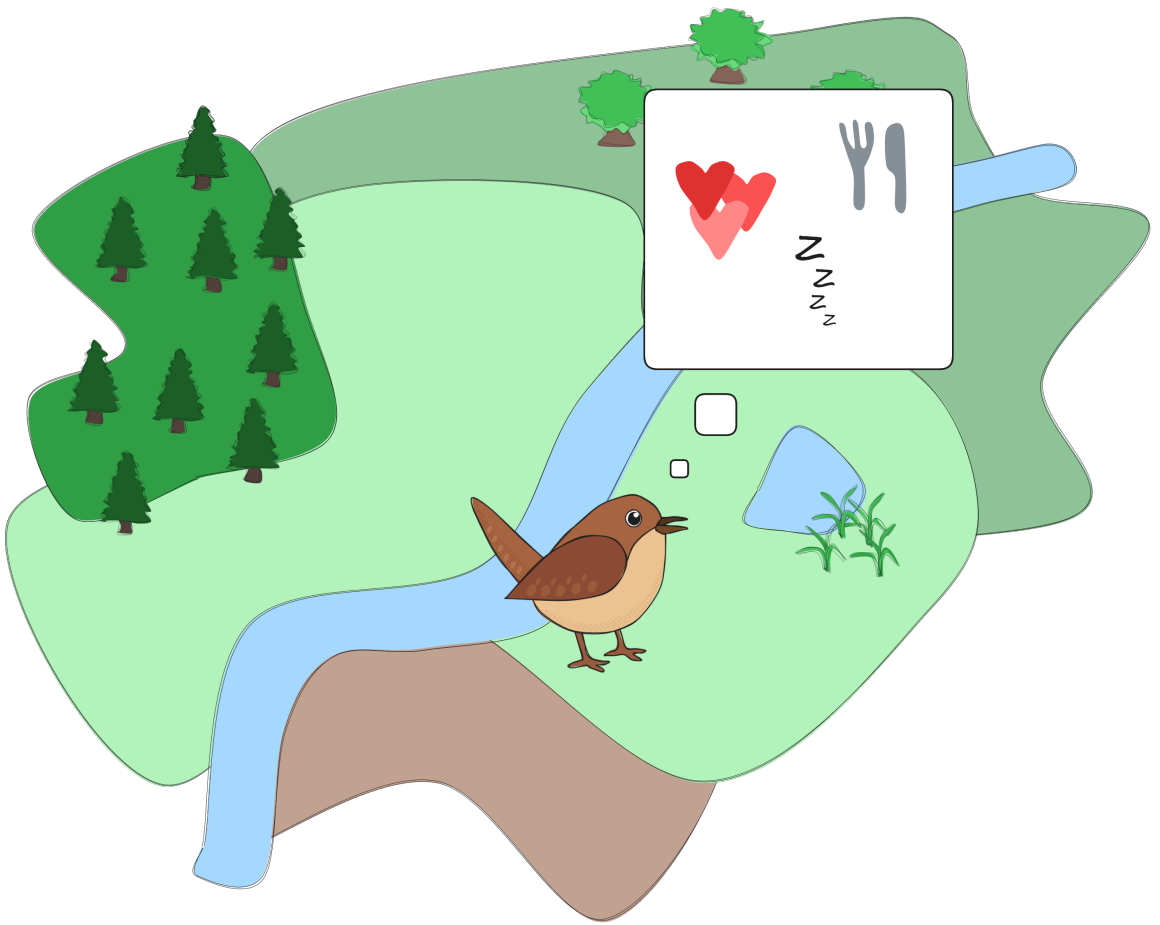

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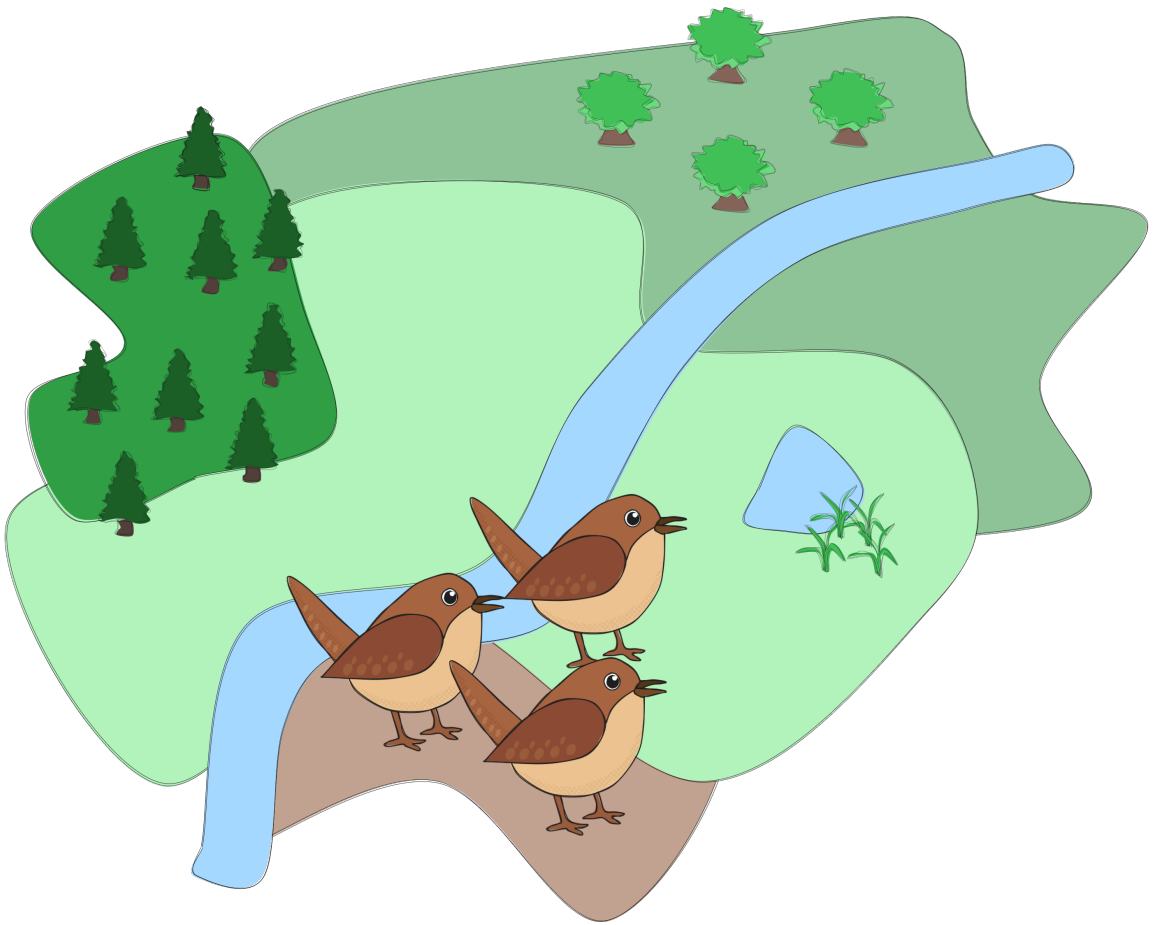

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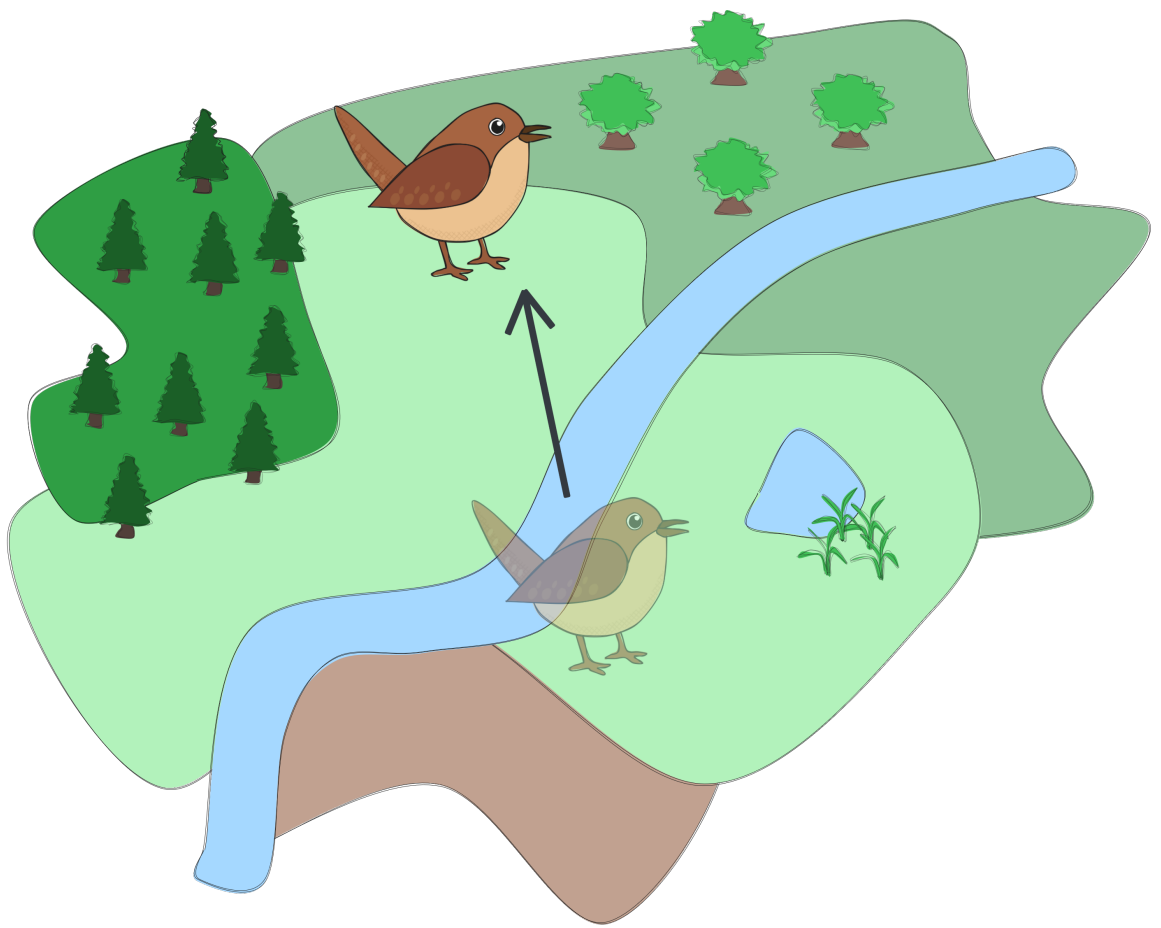

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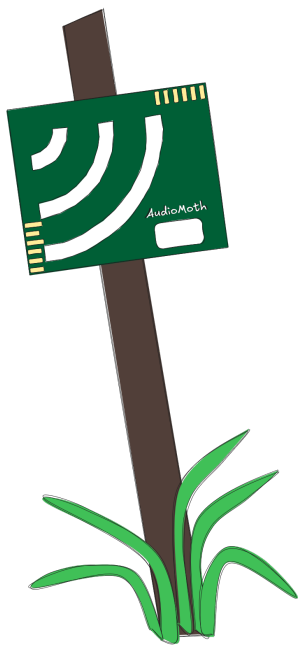
Passive acoustic monitoring

Gibb et al. (2019), Sugai et al. (2019)

Passive acoustic monitoring



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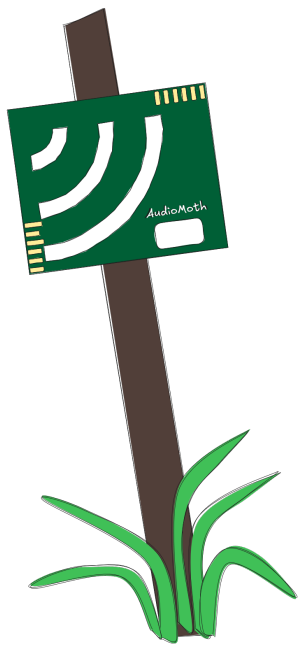

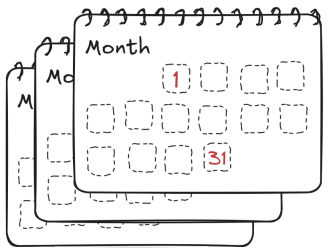

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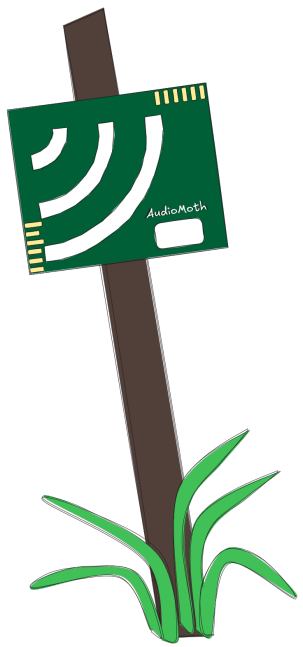

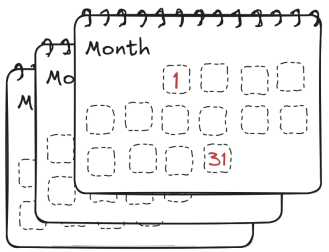

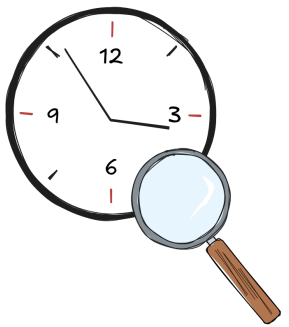

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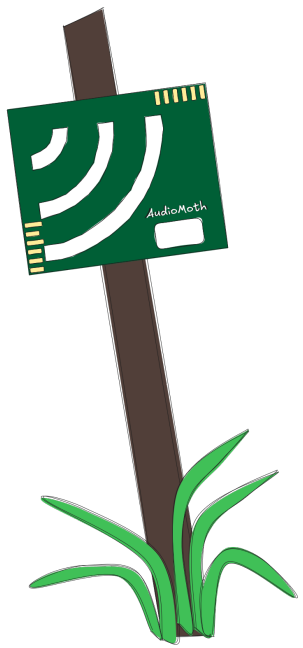
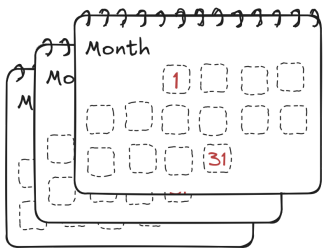
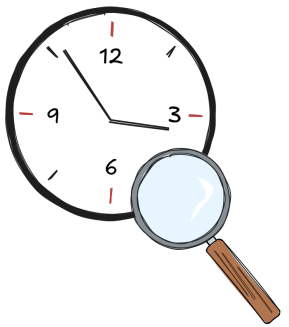
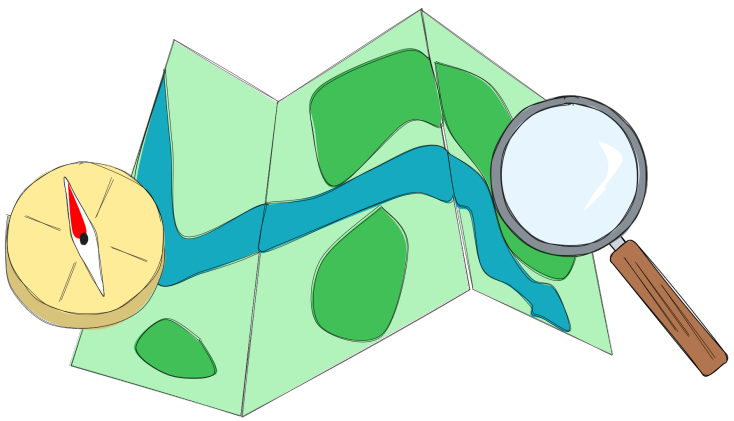








Gibb et al. (2019), Sugai et al. (2019)

Localizing sounds

On the field

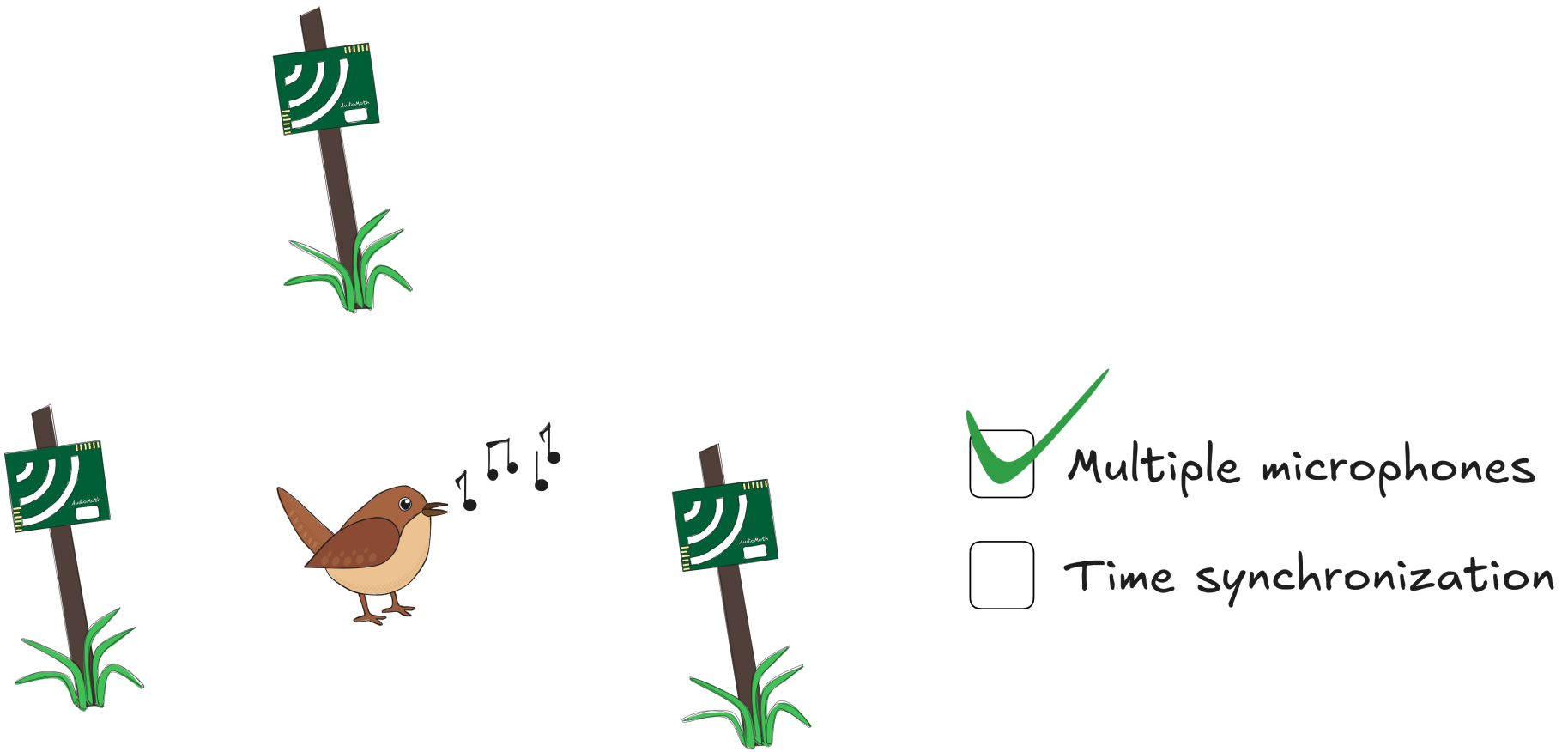
Rhinehart et al. (2020)

On the field

- [ ] Multiple microphones
- [ ] Time synchronization

Rhinehart et al. (2020)

On the field



Rhinehart et al. (2020)

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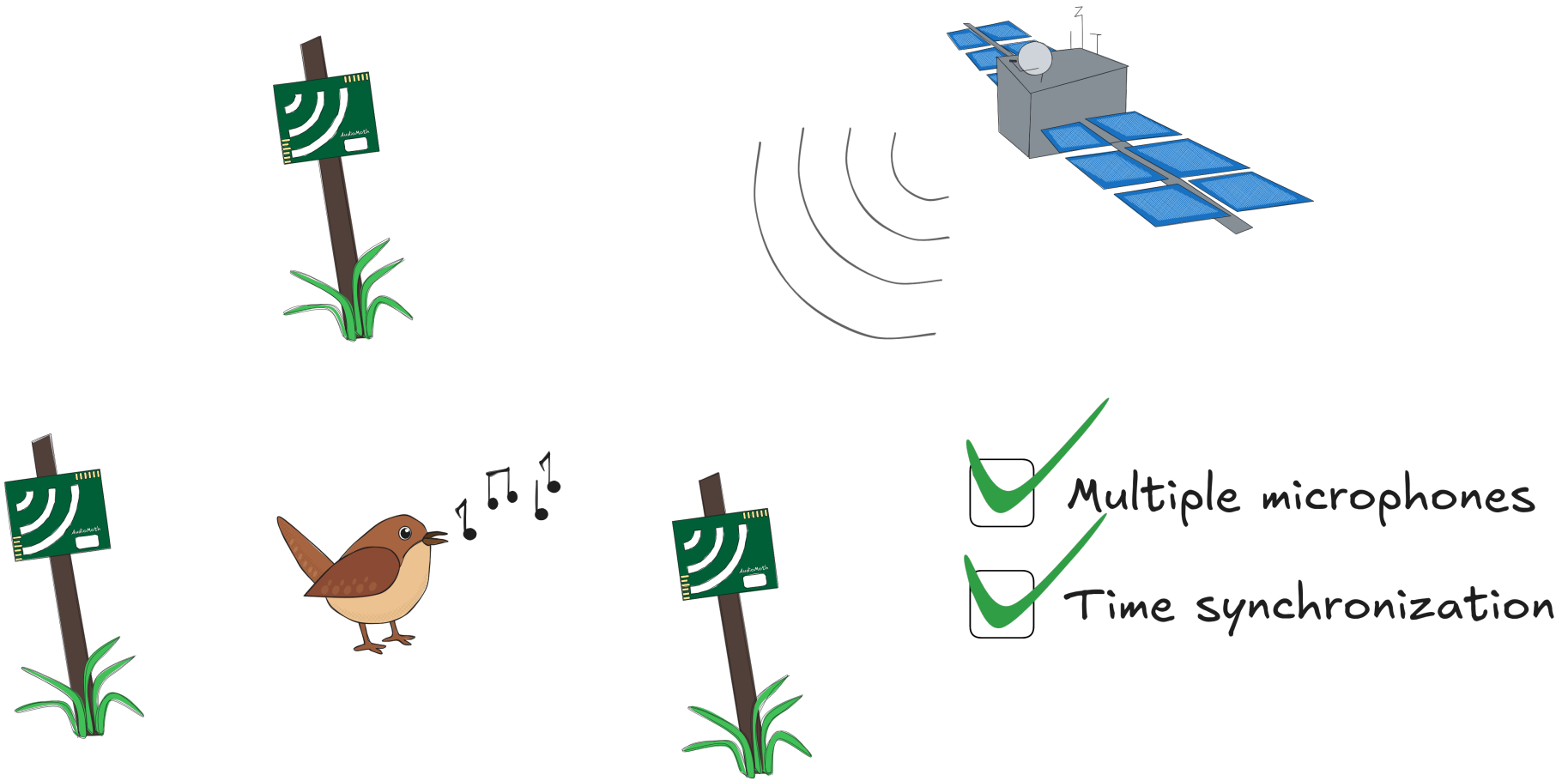

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Objectives

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Localize with precision animal sounds

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Localize with precision animal sounds

Account for uncertainties in:

- Recorders positions
- Time delay estimations

At the lab

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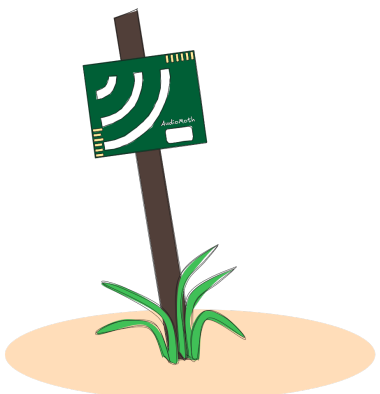

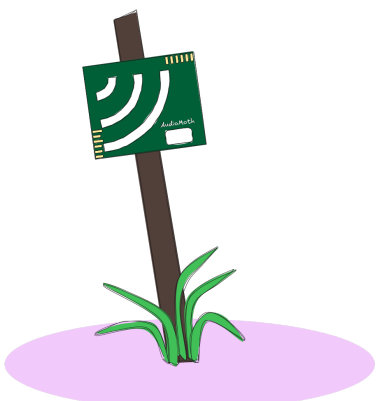

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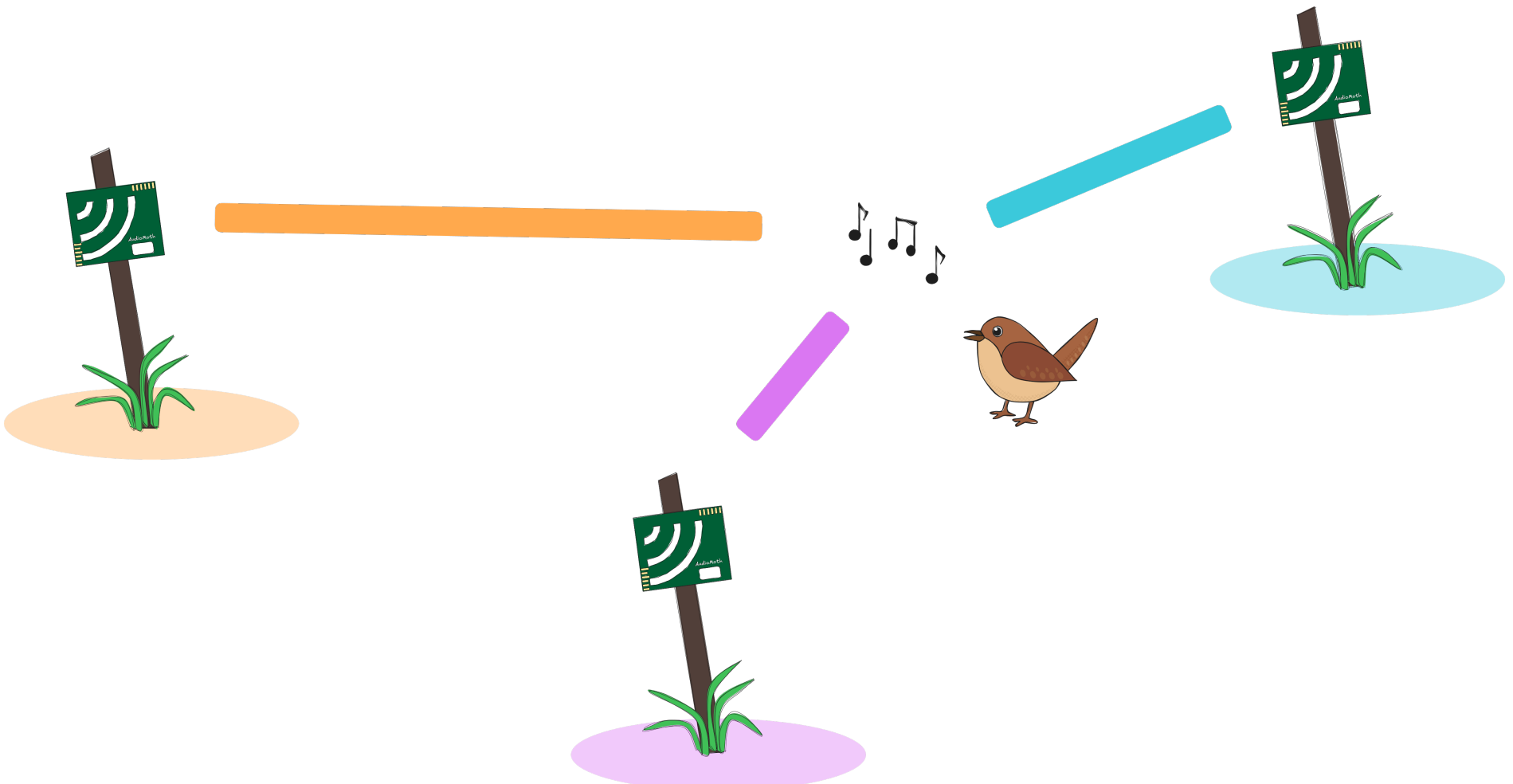

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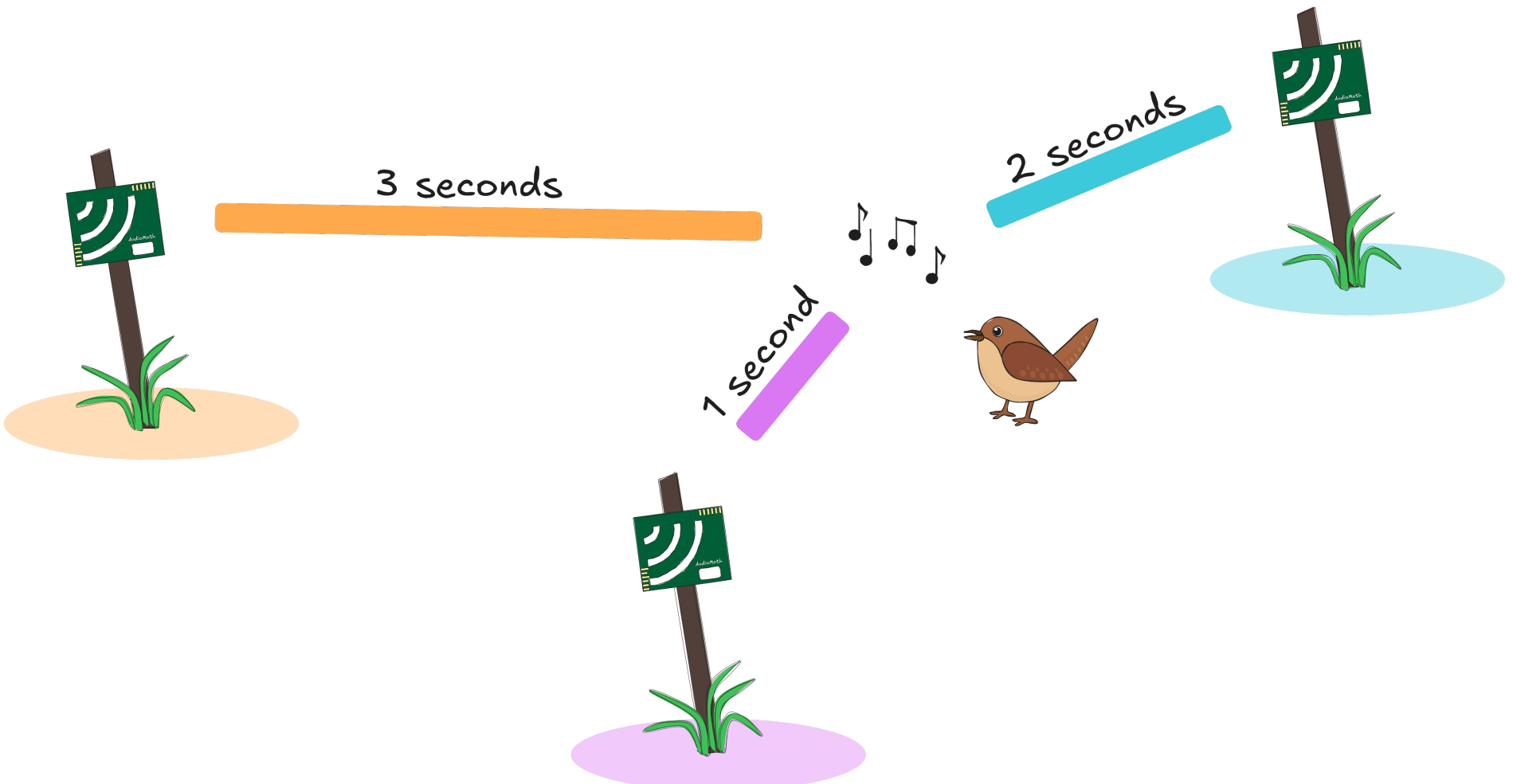

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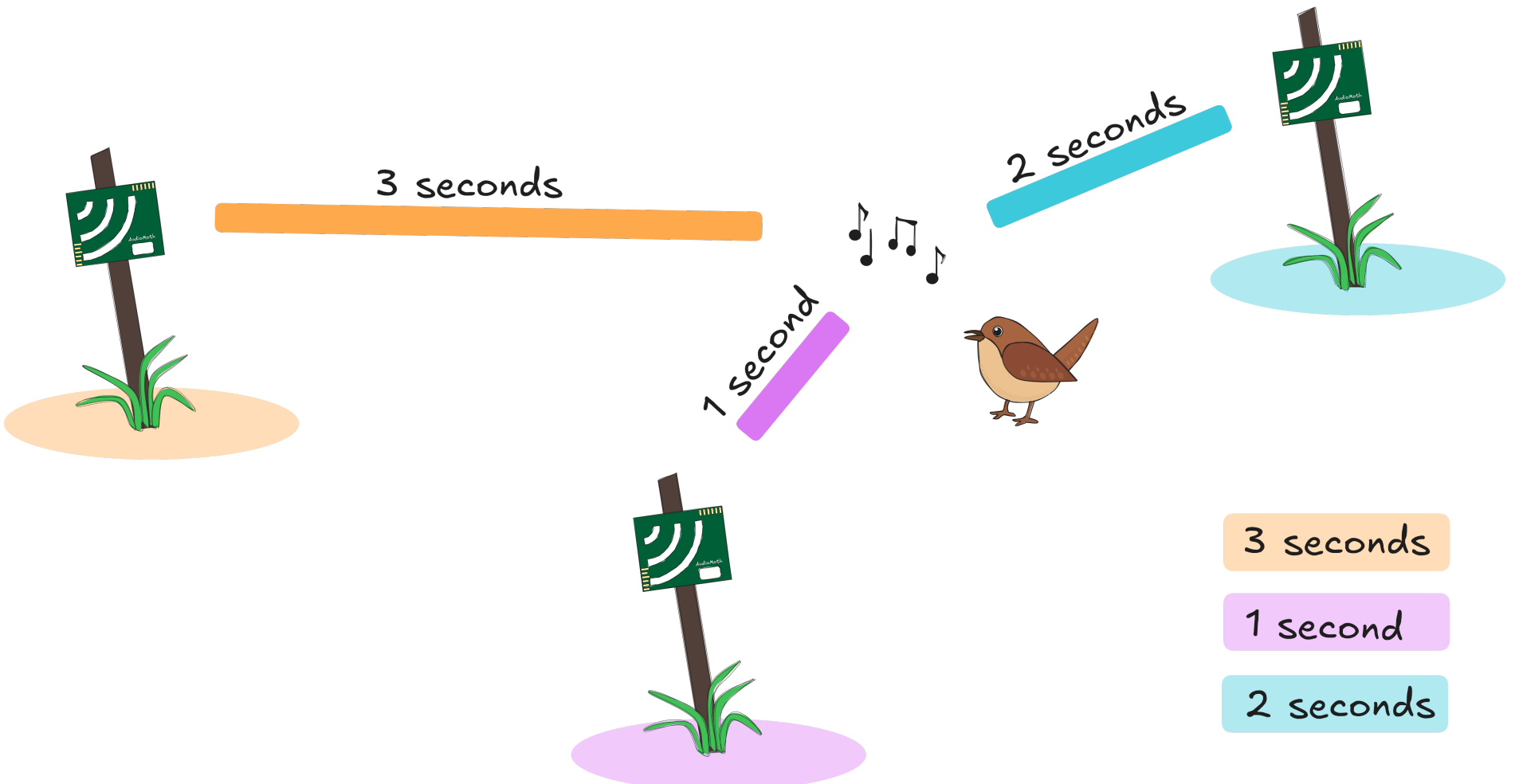

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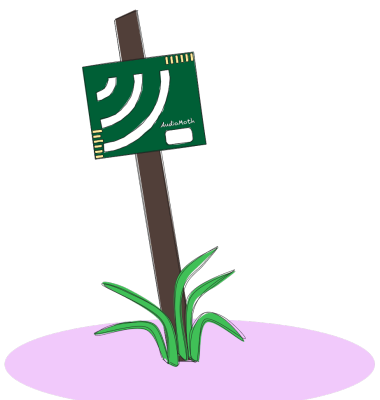

3 seconds

1 second

2 seconds

At the lab

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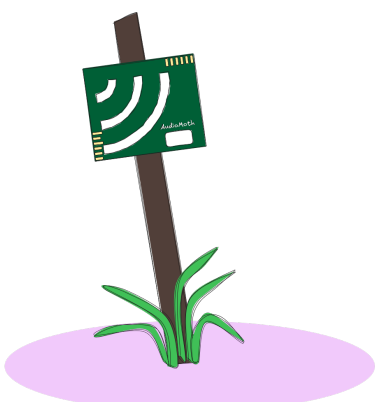

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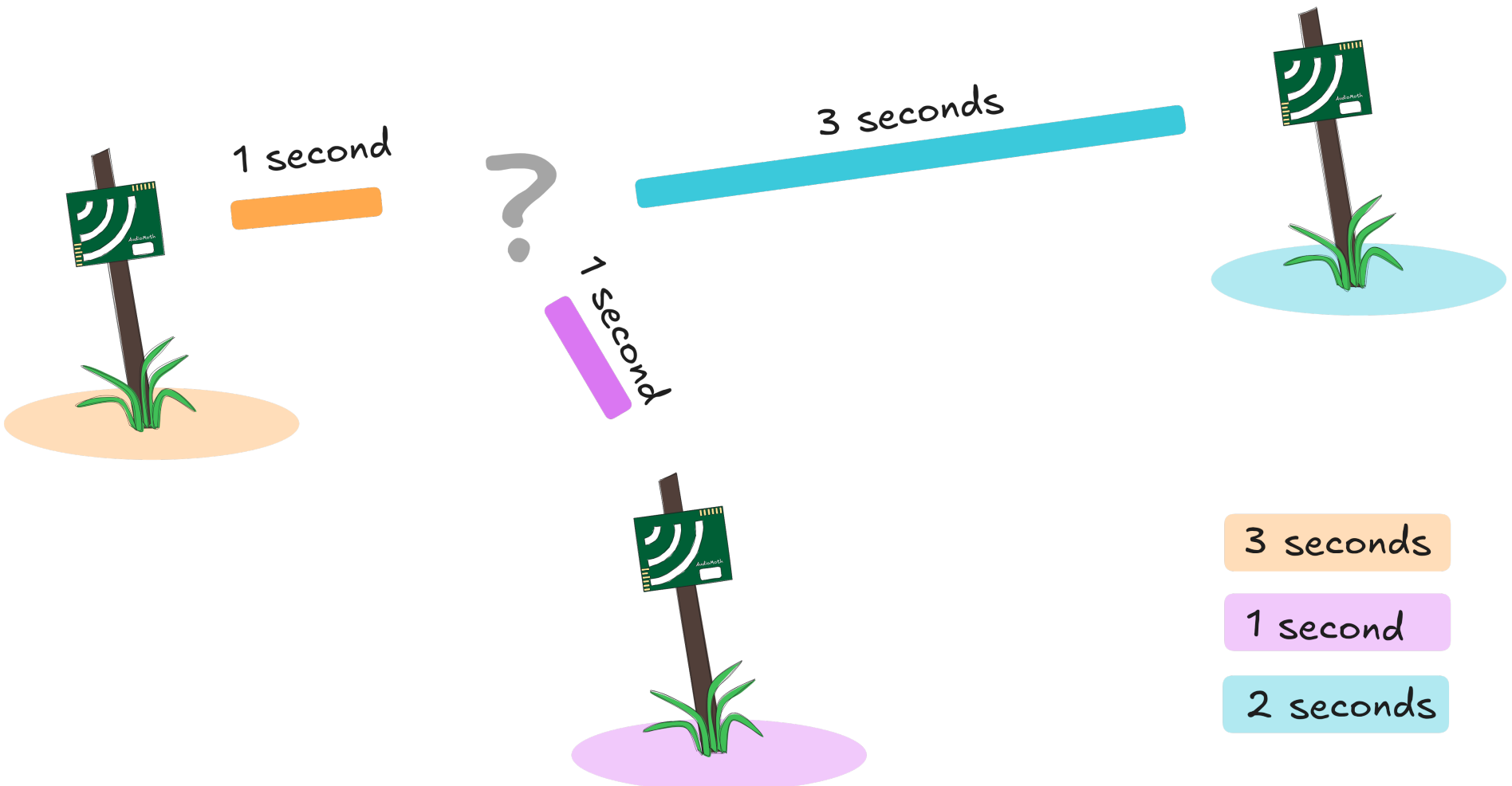

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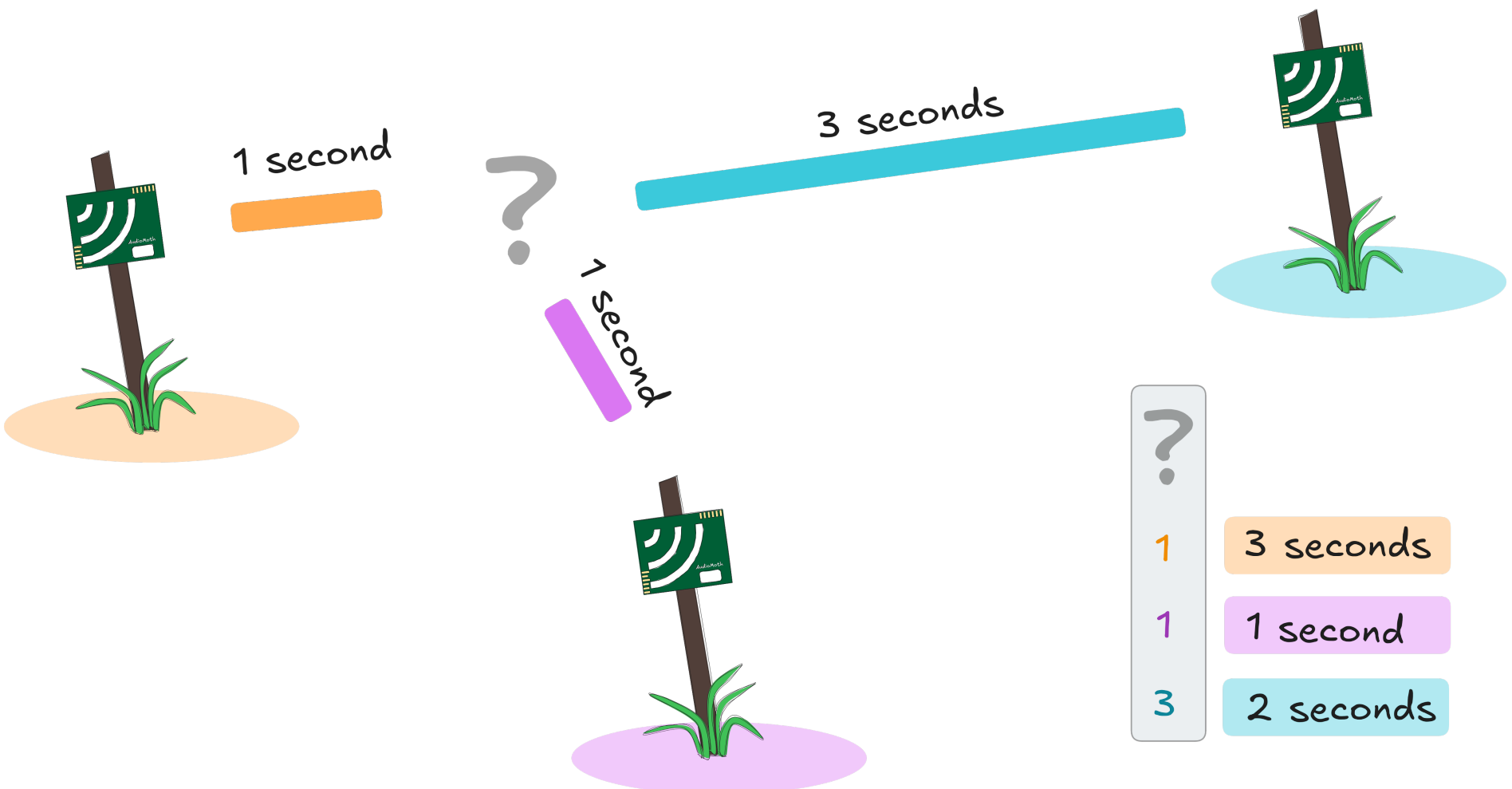

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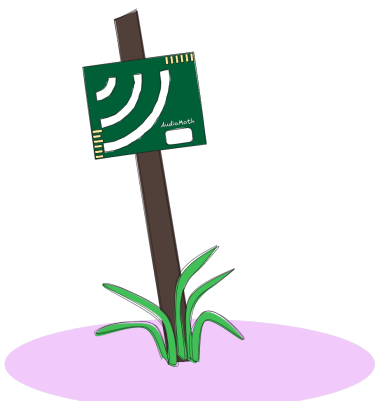

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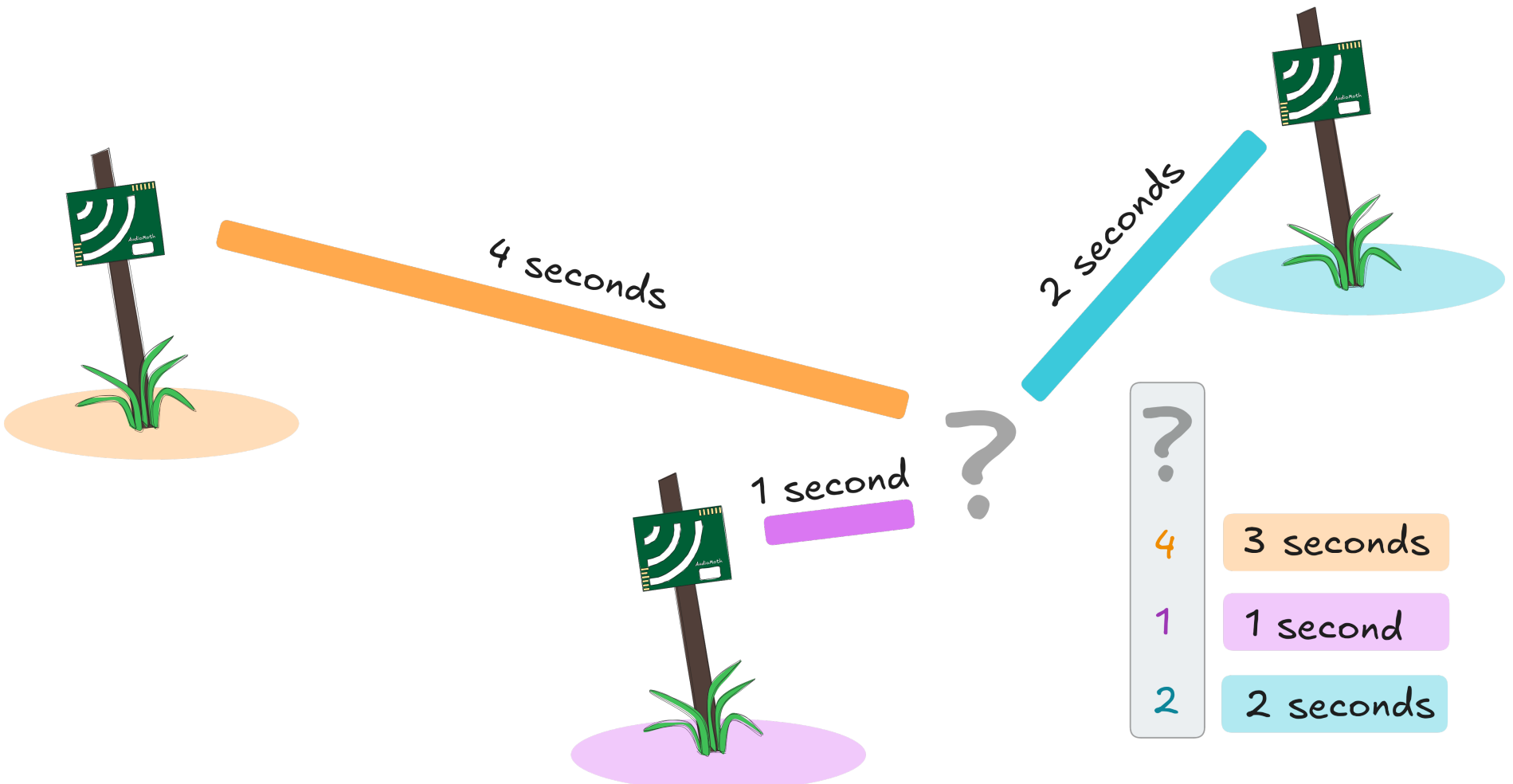

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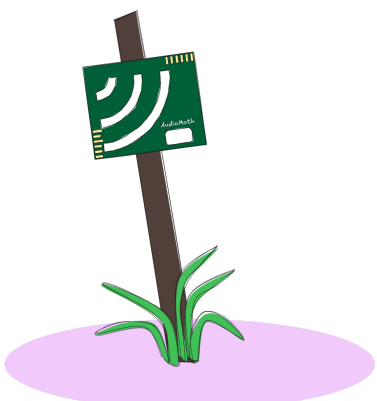

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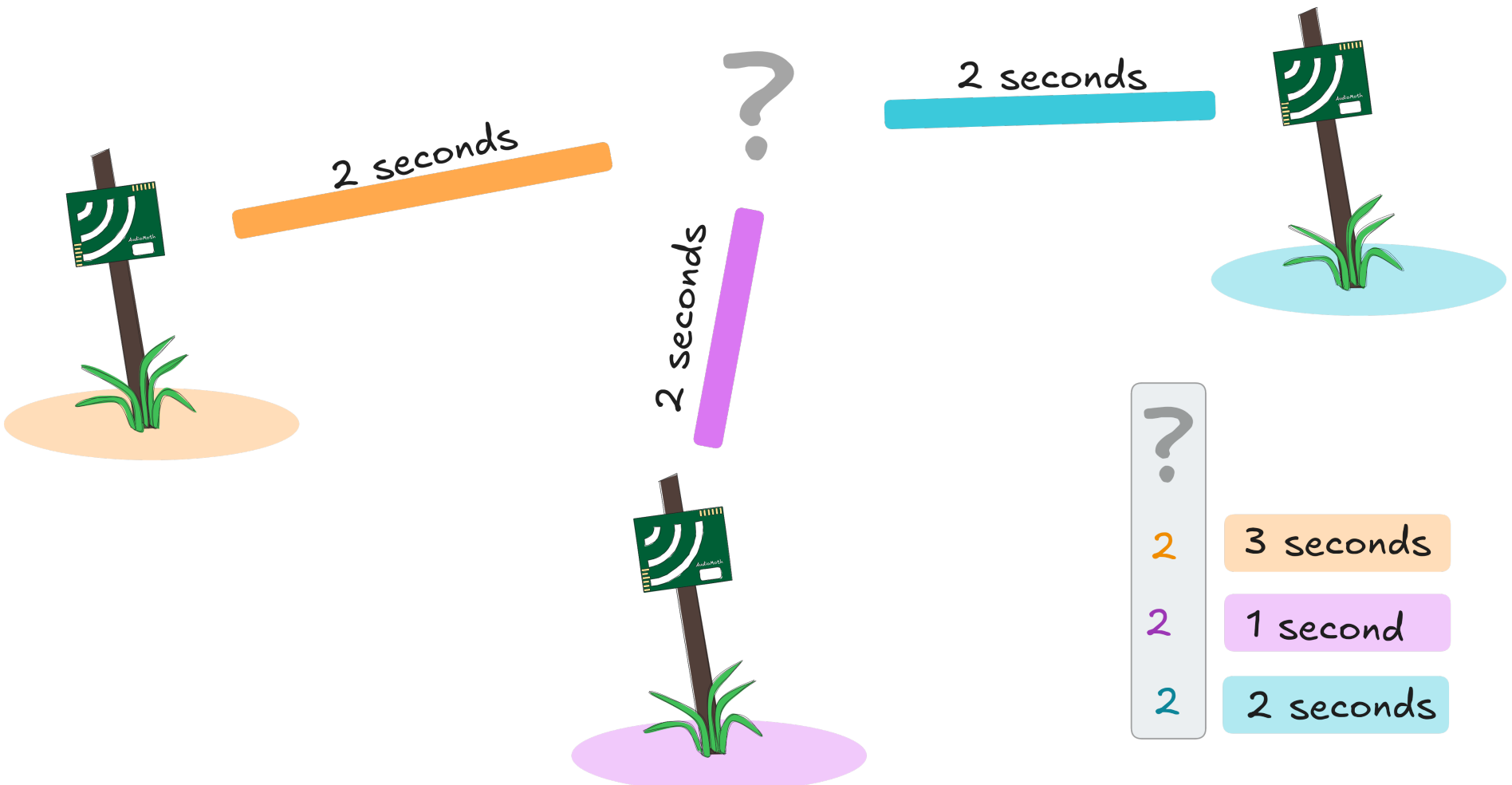
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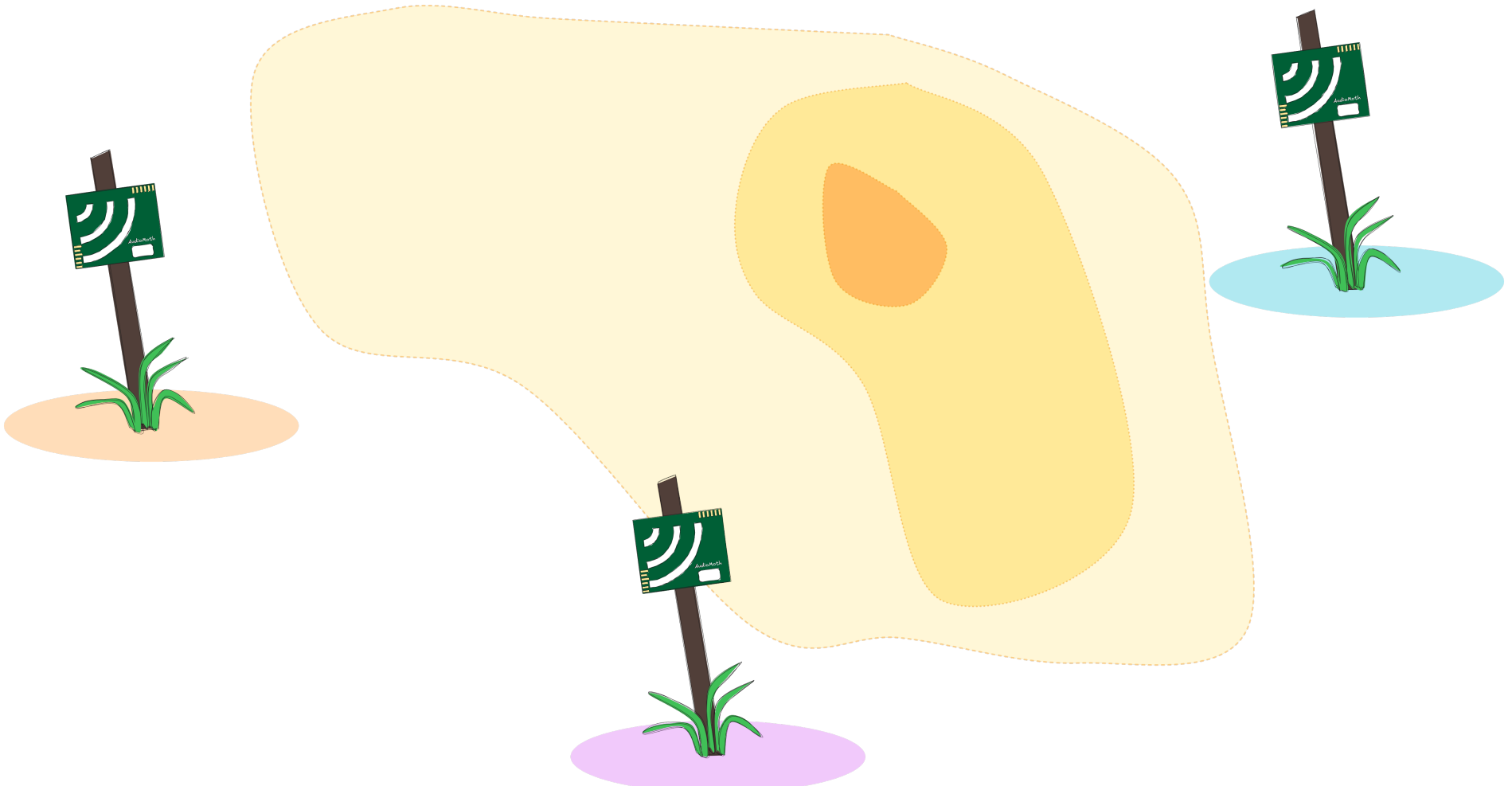

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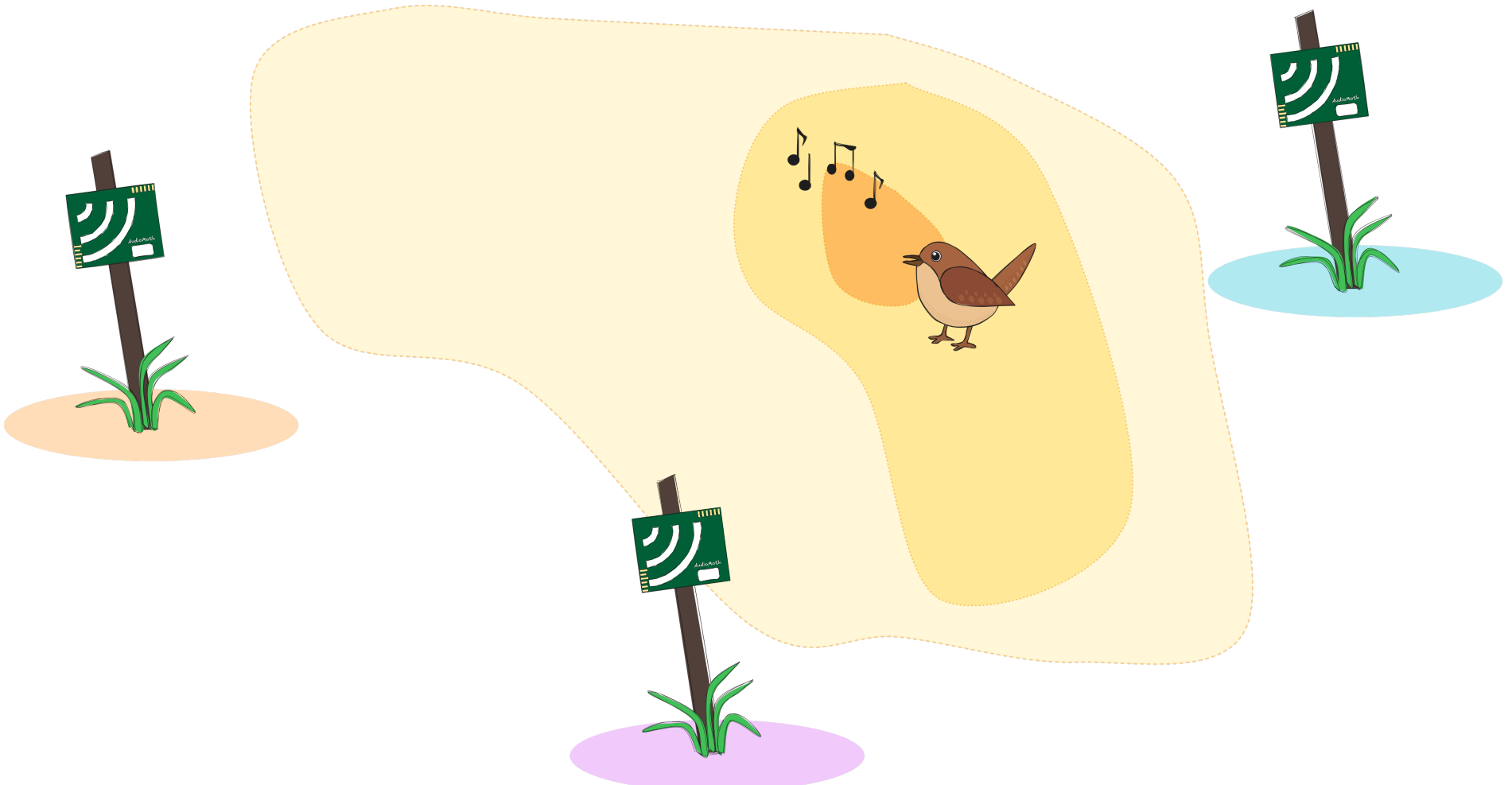

Sound source localization

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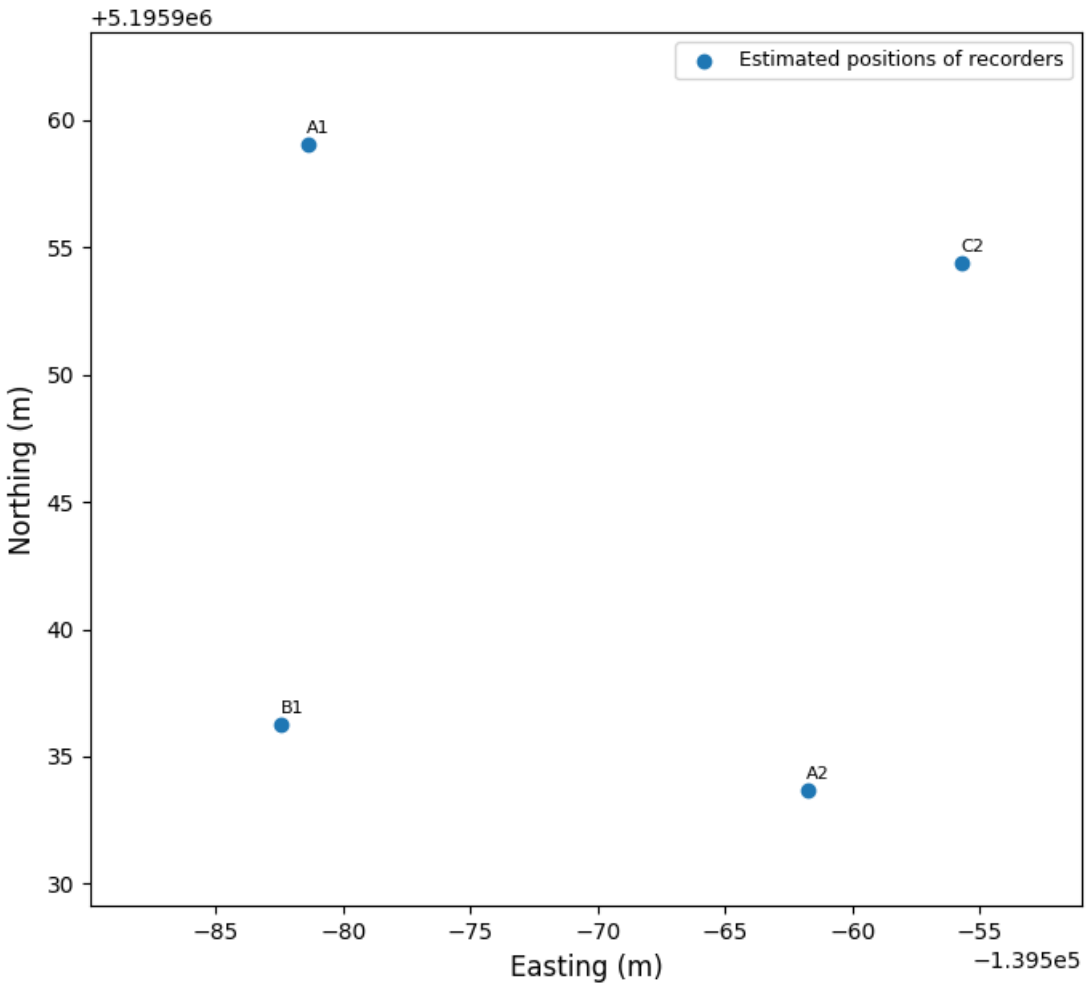

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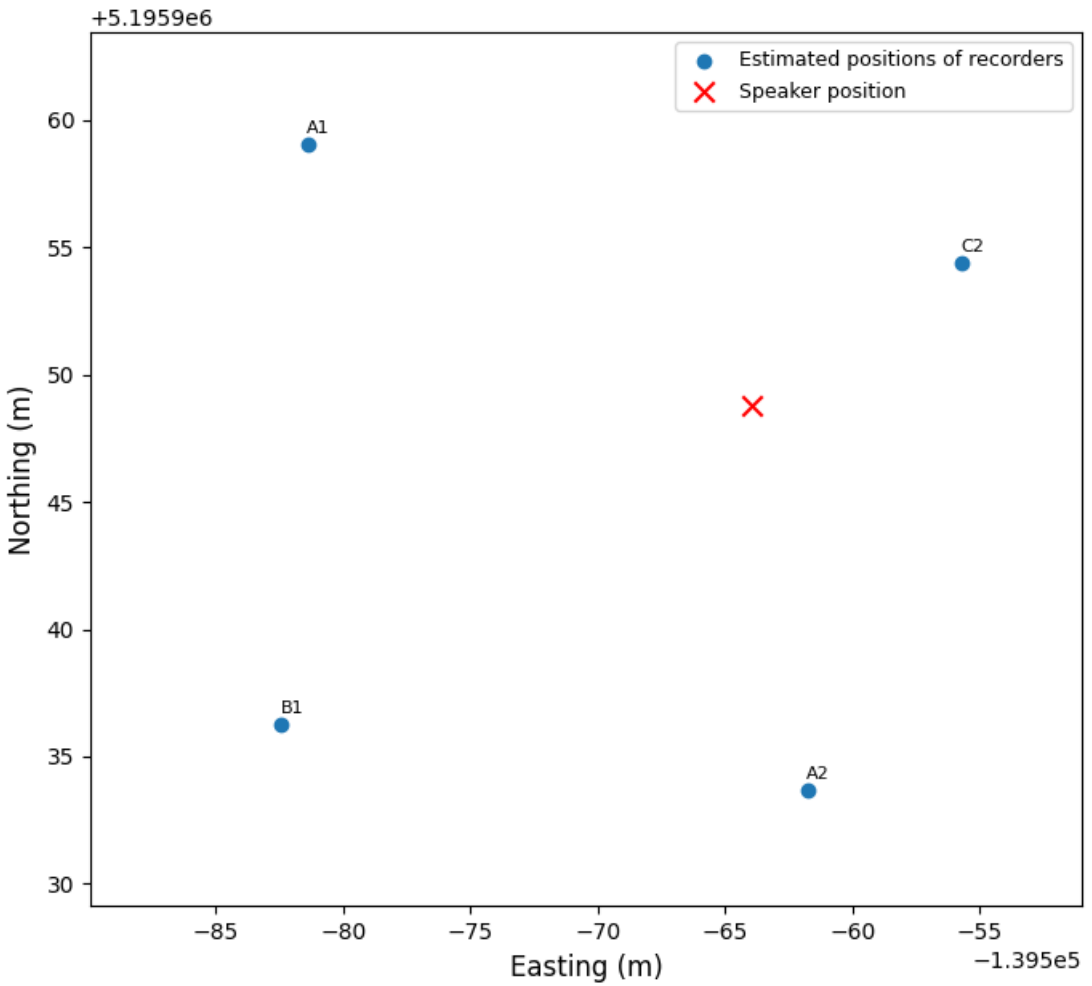

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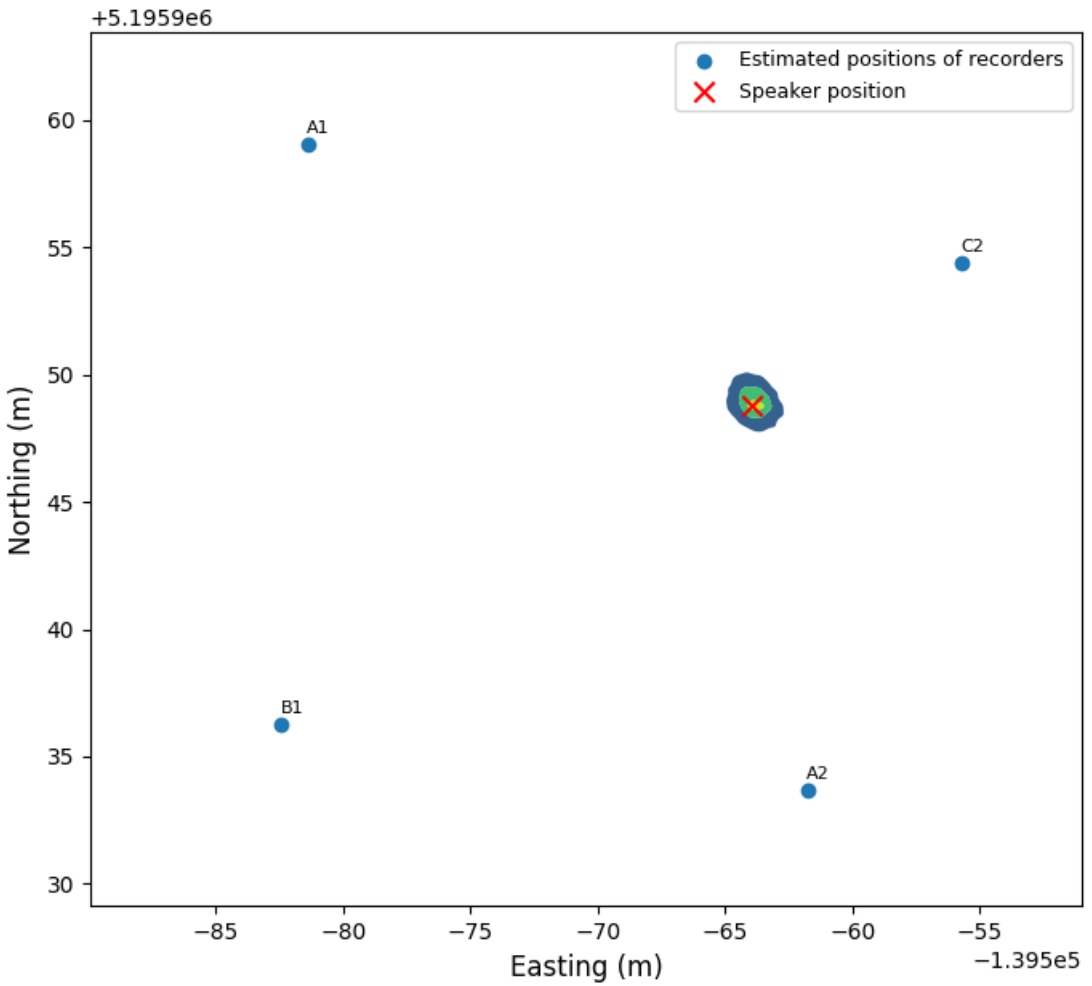

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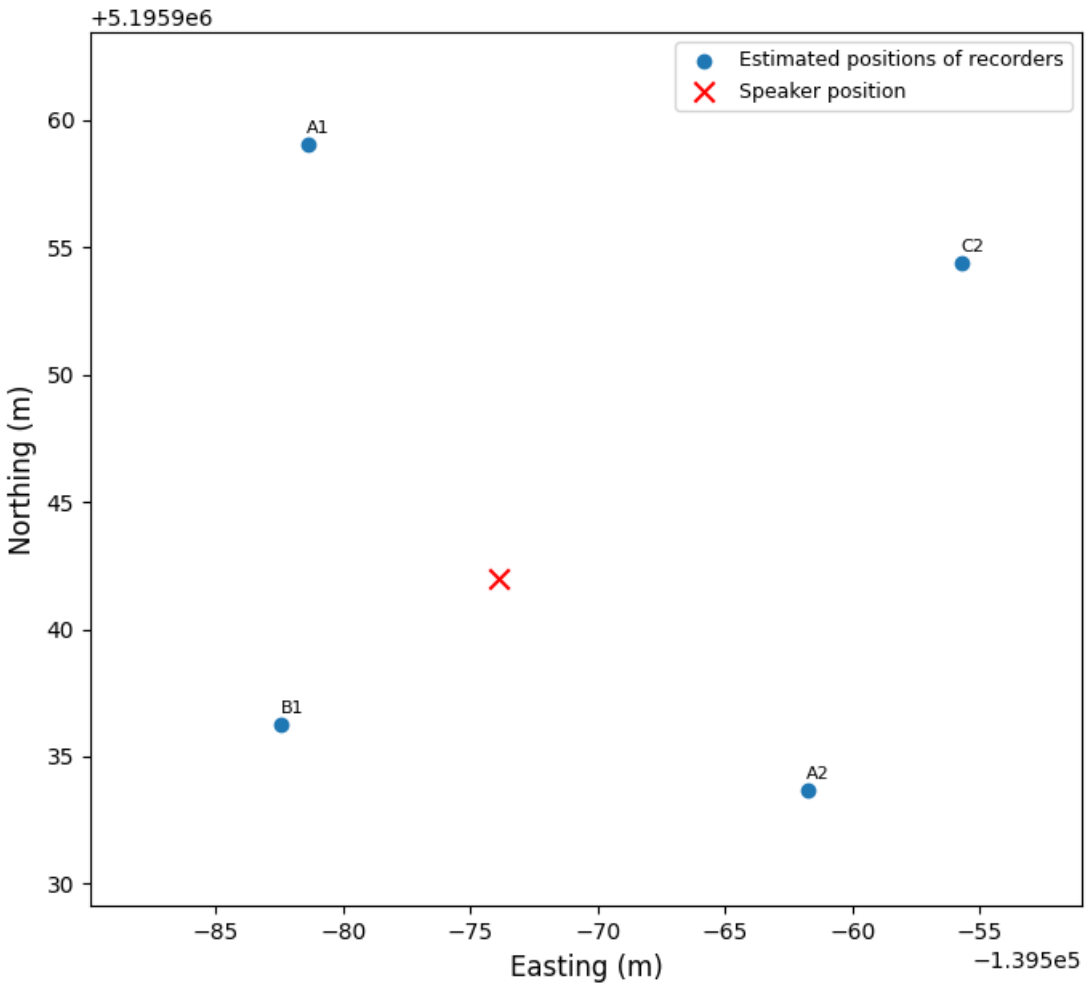

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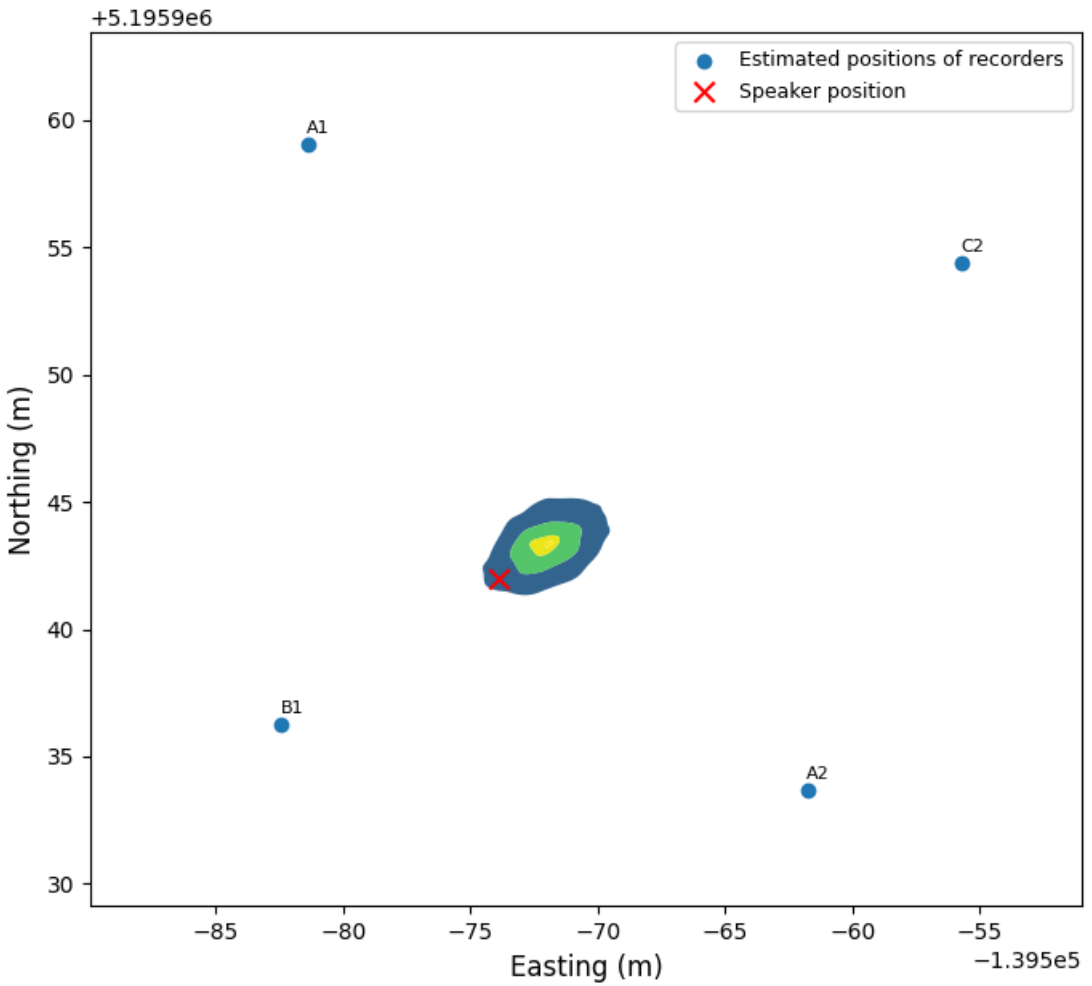

Assessing precision

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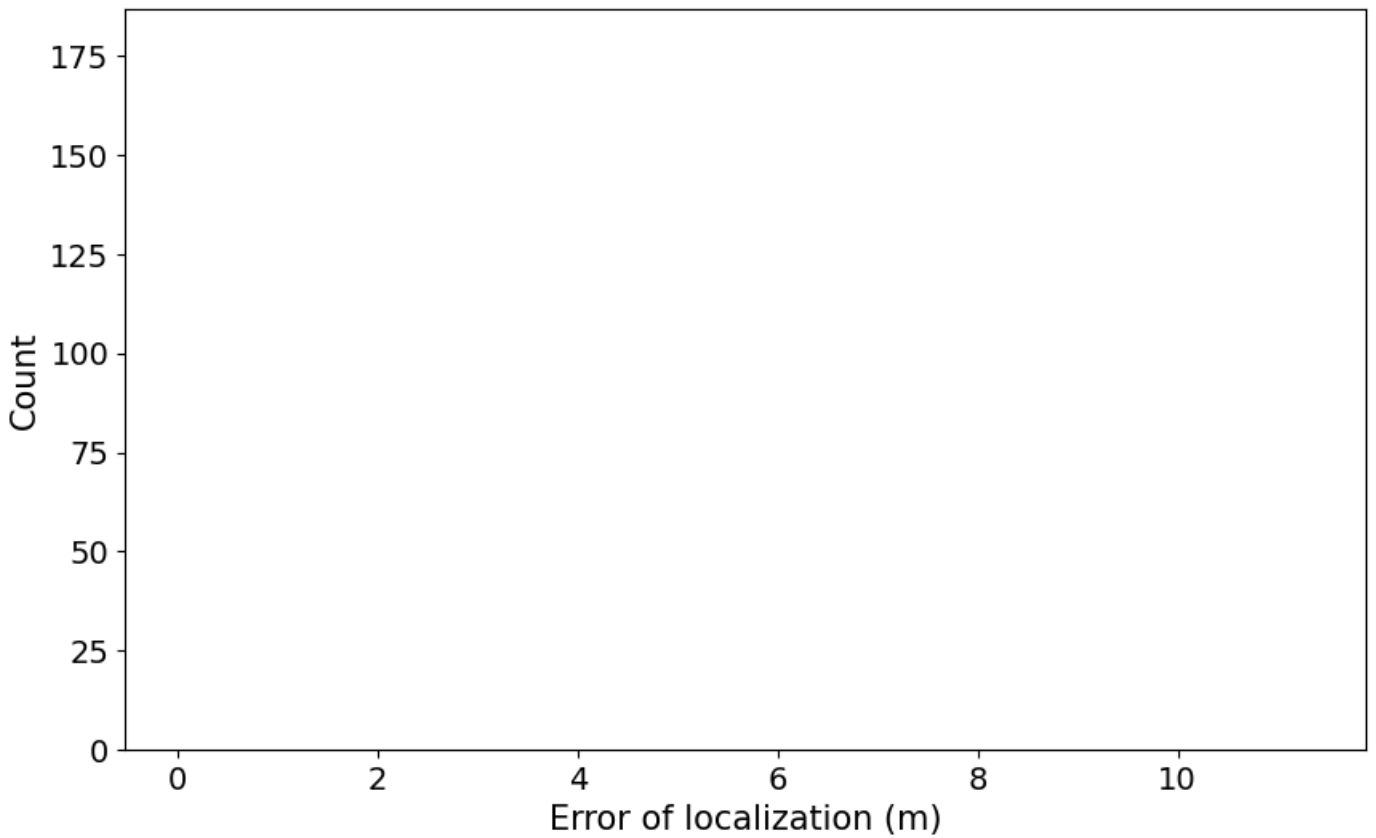

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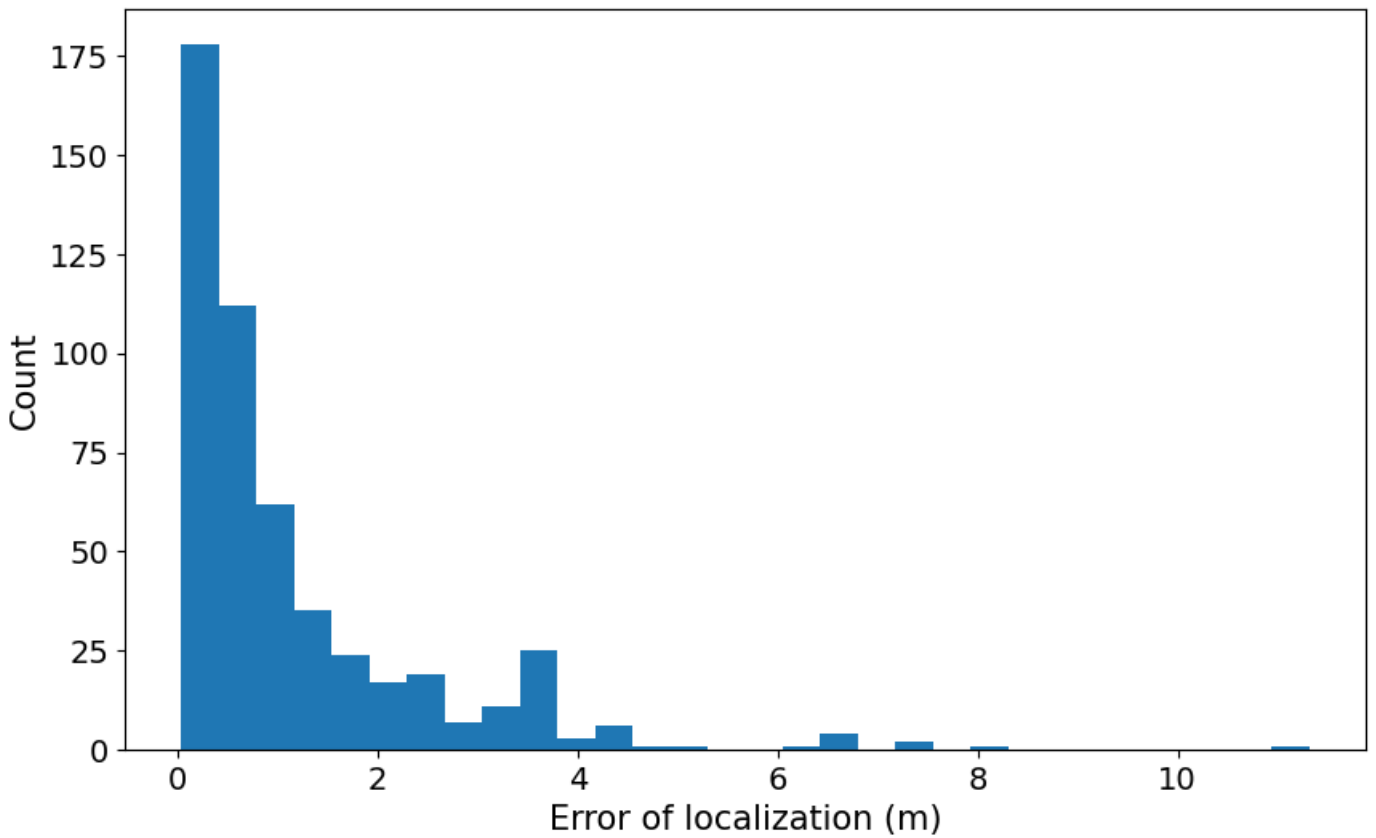

Localizing recorders

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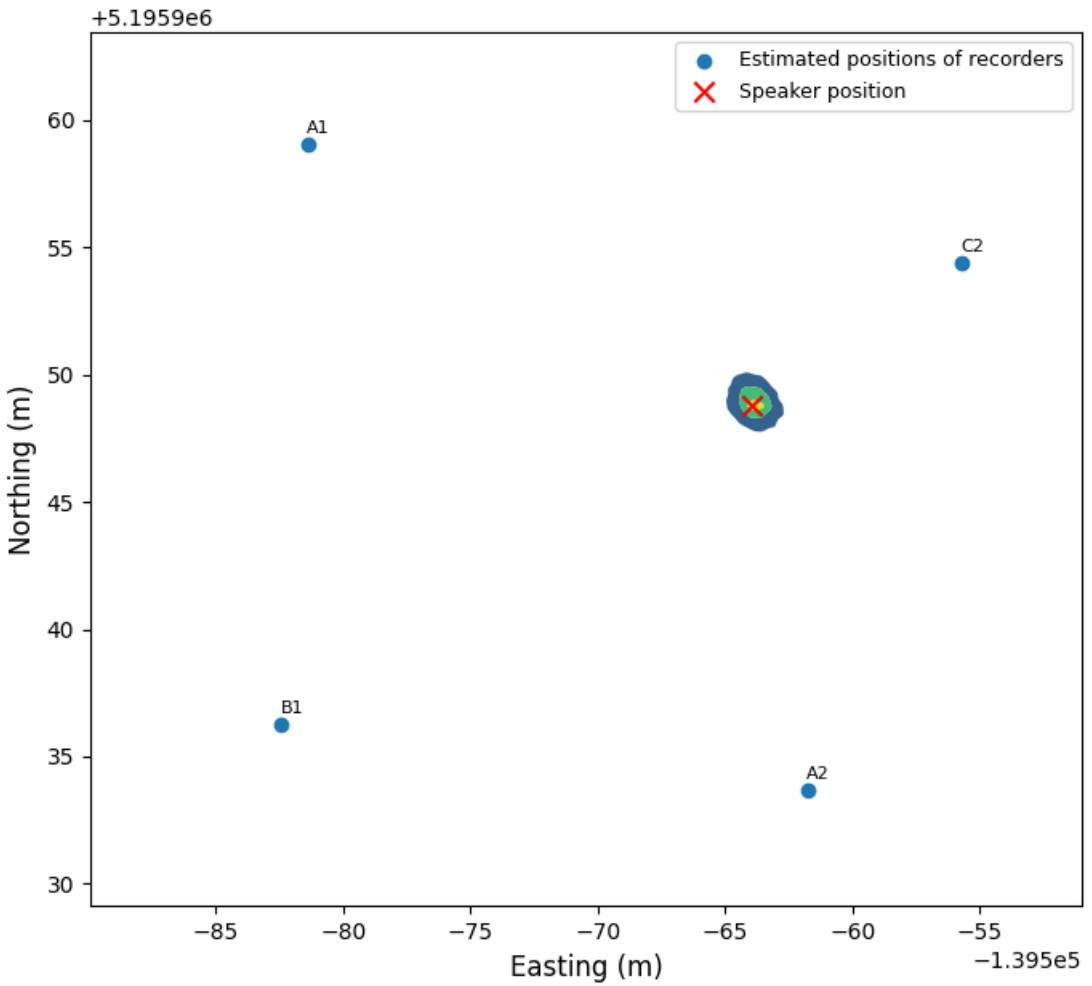

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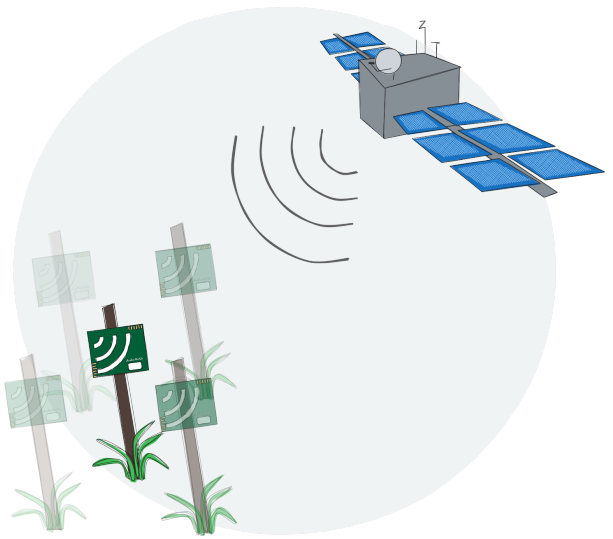

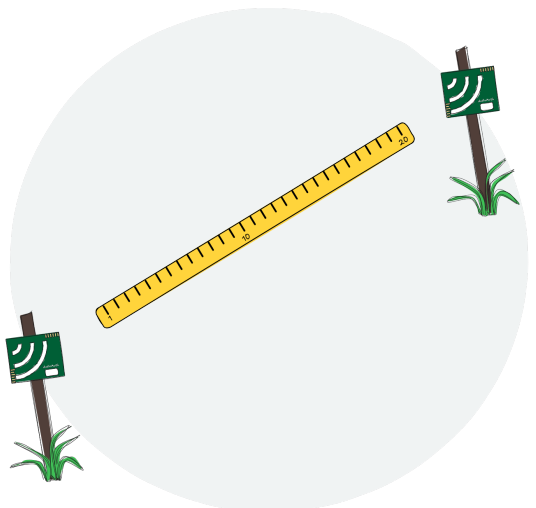

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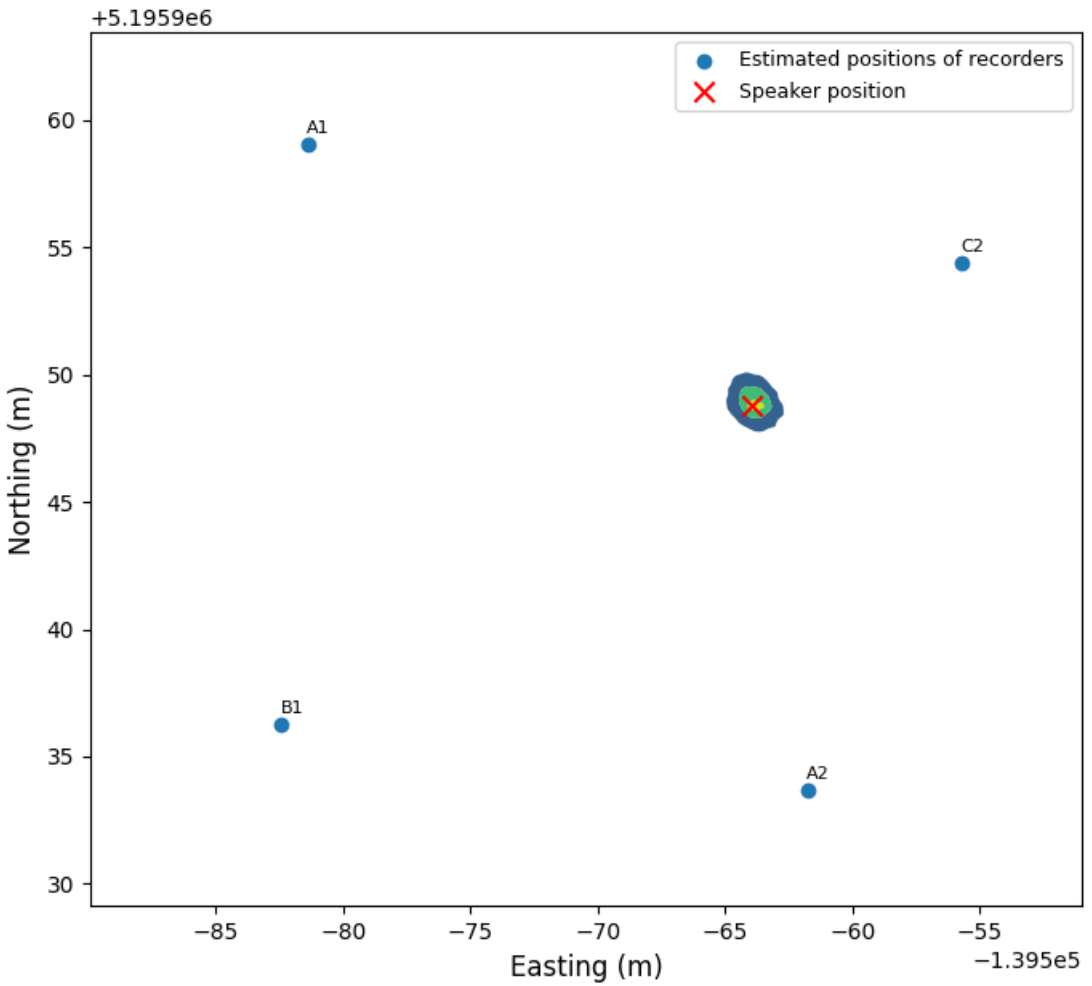

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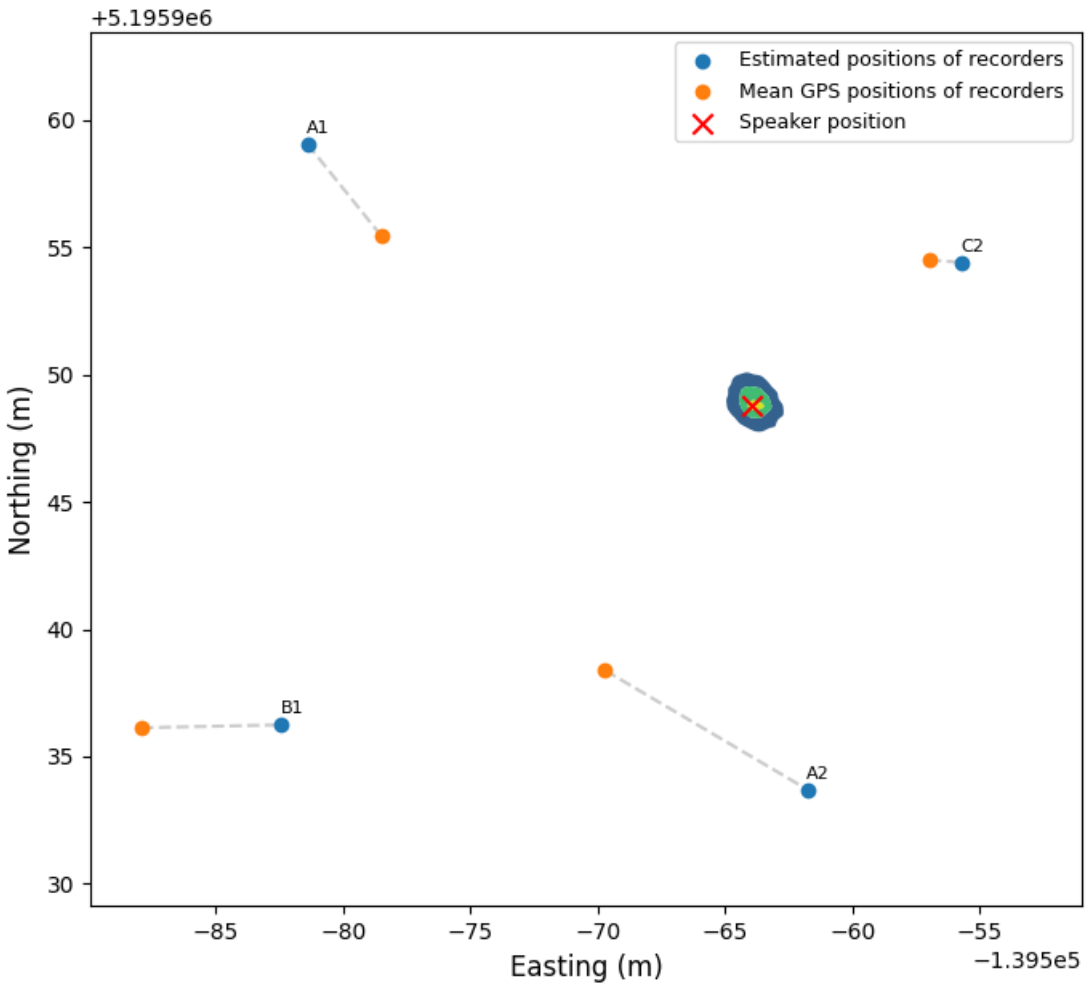

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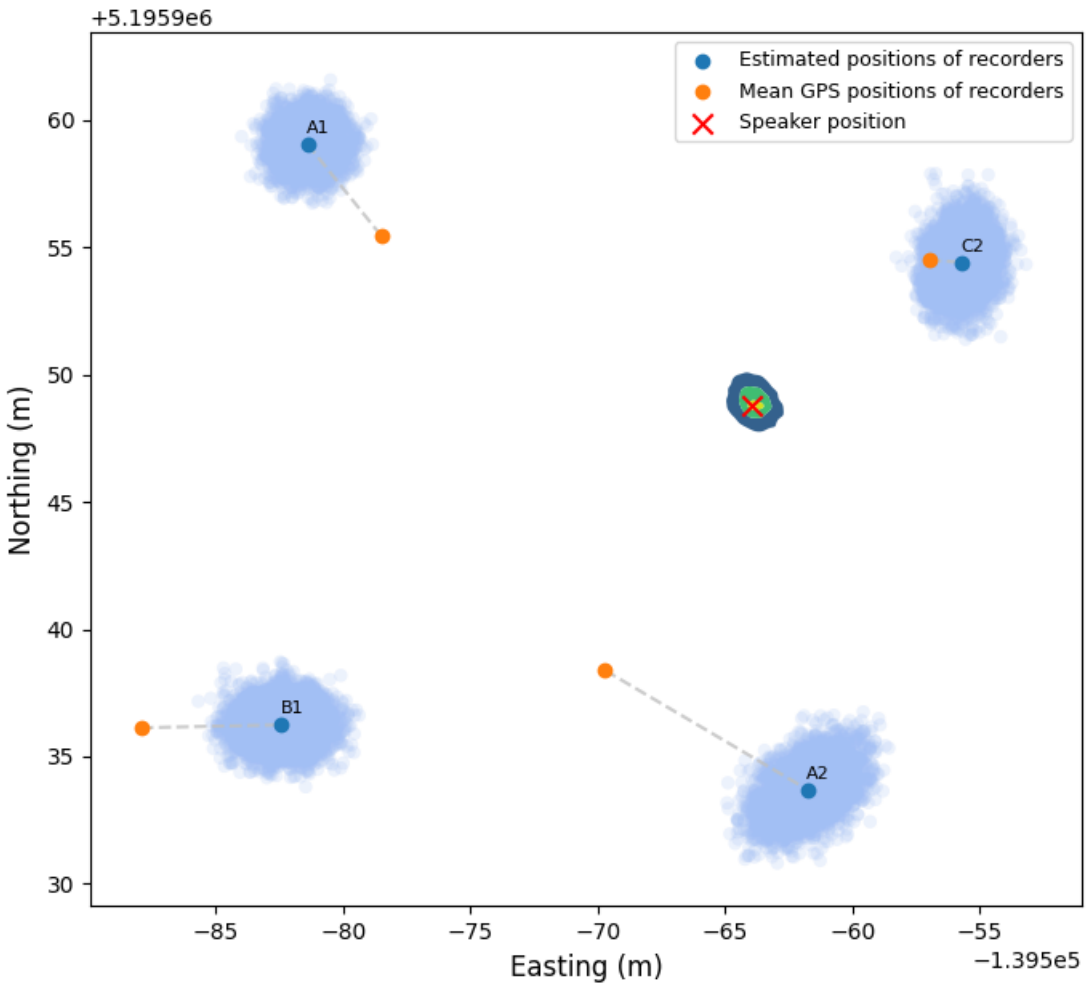

Mapping behaviors

Objectives

✓ Localize with precision animal sounds

Account for uncertainties in:

- ✓ Recorders positions
- ✓ Time delay estimations

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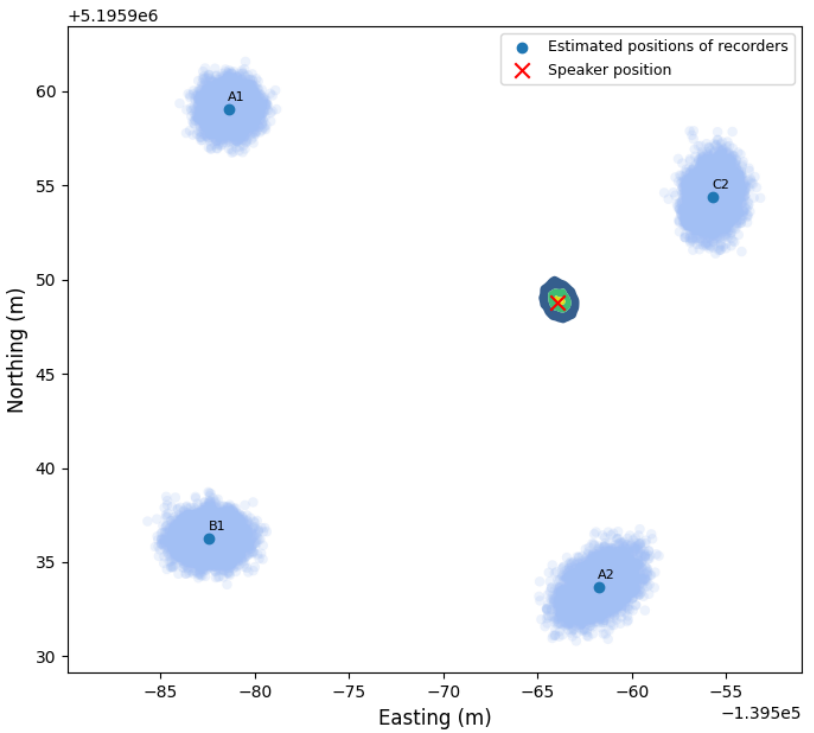

Linking sounds to landscape

Thank you!

This work has been funded by

Contact

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